AI-POWERED SMART ORCHID CARE SYSTEM FOR VANDA CULTIVATION

INDIVIDUAL THESIS

Wickramasinghe D. P.

(IT22065230)

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Student name: Wickramasinghe D. P.

Student ID: IT22065230

Signature:  Date: 10/10/2026

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Role and name	Signature	Date
Ms. Hansi De Silva		
Ms. Thilini Jayalath		

ABSTRACT

Vanda orchids are propagated commercially by hand pollination, and a seed capsule takes roughly nine months to mature. A breeder who pollinates an unfit plant, or who pairs two parents that have never produced a viable cross, learns of the loss only at the end of that period. Both decisions are currently made by eye and from personal experience. This component asks whether pollination readiness can be decided from a photograph alone, without the grower first declaring the plant's condition, and whether cross compatibility can be resolved from the parent names already written on nursery tags. A dataset of 1,190 photographs was assembled, of which 357 whole-plant images from 28 physical plants carry a three-class readiness label. Images are represented by a frozen ResNet18 embedding and classified by XGBoost over 66 image and trait features. Because the 357 photographs come from only 28 plants, validation uses StratifiedGroupKFold grouped by plant; an earlier image-level split had reported perfect accuracy, which grouping exposed as leakage. The deployed model reaches 0.717 accuracy and 0.715 macro F1 against a 0.639 majority-class baseline, recovering the Not Suitable class at 98 per cent. A two-stage input gate screens every photograph before a verdict is issued: a one-class k-nearest-neighbour test over the embedding refuses images whose cosine distance to their tenth neighbour exceeds 0.25, and a logistic filter separates orchid blooms from other flowers. On a held-out validation set the gate refuses 91.9 per cent of non-orchid images, including all 309 images in seven non-plant categories, at an area under the curve of 0.921 and a false-refusal rate of 2.2 per cent. Cross compatibility is reported as an evidence tier over registered precedents, never as a success percentage, because the register records only crosses that succeeded and therefore has no denominator.

Keywords: Vanda orchid, pollination readiness, transfer learning, out-of-distribution detection, grouped cross-validation

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I am grateful to the orchid growers who allowed me into their shade houses and let me photograph their plants over several months, and who answered a great many questions about how they decide which plant to breed from. I owe a particular debt to the breeder at Peradeniya whose account of intergeneric crossing contradicted the published compilations I had been reading. That contradiction prevented me from reporting a success rate the available data could never have supported, and it shaped the design of the whole second level of this component.

I thank my three teammates on project R26-SE-018 for the shared backend and database that made integration straightforward, and in particular the owner of Component 2, whose disease classifier this component consumes at run time. Finally, I thank the evaluation panel at the progress presentations, whose questions about input validation and about what a register of successes can support led directly to the two-stage gate of Section 3.3 and the evidence tiers described there.

This project received no external funding. Photography was carried out with my own equipment, and the models were trained on departmental and personal hardware.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Network
CV	Cross-Validation
ExG	Excess Green Index
F1	Harmonic Mean of Precision and Recall
HSV	Hue, Saturation, Value colour space
JSON	JavaScript Object Notation
k-NN	k-Nearest Neighbours
OOD	Out-of-Distribution
RBF	Radial Basis Function
REST	Representational State Transfer
RGB	Red, Green, Blue colour space
RHS	Royal Horticultural Society
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SVM	Support Vector Machine
VPD	Vapour Pressure Deficit

1 INTRODUCTION

1.1 Overall project context

Project R26-SE-018, the AI-Powered Smart Orchid Care System, is an integrated research project carried out by four students. It addresses a single observation about Vanda cultivation in Sri Lanka: the decisions that keep a plant productive are well understood by experienced growers, documented in the horticultural literature, and nevertheless taken by eye, from memory, one plant at a time. The project partitions those decisions into four and gives each one a measurement.

The four components are hybrid pollination and compatibility analysis, which is the subject of this thesis; disease detection and severity grading; weather-driven irrigation and fertilisation; and growth-stage classification with bloom timing. They share one FastAPI backend, one Firebase realtime database and one React Native application, so a verdict produced by one component is visible to the others and to the grower in a single record per plant.

This thesis concerns Component 1 only. The other three are the work of other team members and are described here solely to the extent that this component depends on them. The integrated argument, the shared evaluation framework and the group-level results are presented in the common report submitted alongside the four individual theses.

1.2 Individual component context

Component 1 answers two questions that a breeder asks in sequence. The first is whether a particular plant is in a fit state to be bred from at all. The second is whether a proposed pairing of two named parents has any precedent behind it. The two questions require different kinds of evidence, and this component answers them with different mechanisms: the first from a photograph, the second from the registered hybrid record. Figure 1.1 places the component in the wider system.

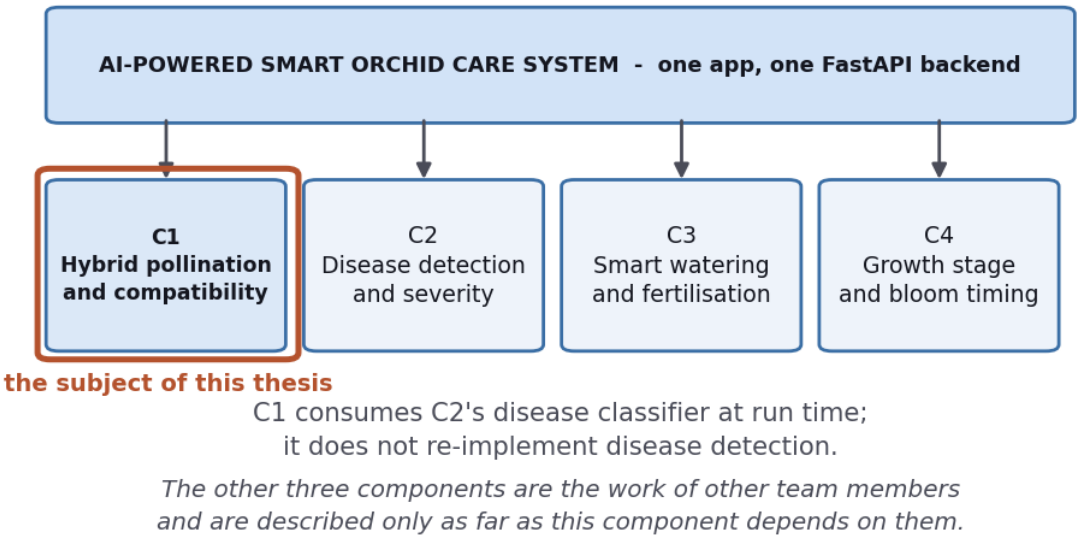


Figure 1.1: Component 1 within the integrated solution. The component consumes Component 2's disease classifier at run time rather than training a second disease model of its own

The separation matters because it is not arbitrary. Whether a plant is healthy enough to carry a capsule is visible in the plant and therefore measurable from an image. Whether two lineages can produce viable offspring is a property of their genetics and their recorded breeding history, and no photograph contains it. Treating both as image problems, which an earlier design briefly did, produces a model that appears to answer the second question while actually answering the first one twice.

1.3 Problem statement

Commercial propagation of named Vanda hybrids is by hand pollination. Pollinia are transferred from a donor flower to the receptive stigma of a pod parent, the ovary swells over several weeks, and a mature capsule is harvested after roughly nine months and sown under sterile conditions. The interval between the decision and its consequence is the defining feature of the task. A grower who pollinates a stressed plant does not find out for most of a year, and by then the conditions that caused the failure cannot be reconstructed.

Two judgements govern the outcome and both are currently made unaided. The first is whether this plant is fit to carry a capsule, which rests on maturity, vigour, root

condition, the absence of active disease and the stage of the flower. The second is whether this pairing is worth attempting, which rests on knowledge of which lineages have crossed before. Neither judgement is recorded, so a grower cannot review a past decision, and neither transfers to a less experienced grower except by apprenticeship.

The research problem addressed by this component is therefore stated as follows. Given a single photograph of a Vanda plant taken under ordinary shade-house conditions, and given the names of two candidate parents as they appear on nursery tags, can a system produce a pollination recommendation that is measurably better than the majority-class baseline, that does not depend on the grower first declaring the plant's condition, that declines to answer when the photograph is not of an orchid, and that states the evidence behind a proposed cross without fabricating a success probability the available data cannot support?

1.4 Research gap

Three gaps were identified in the course of the review reported in Chapter 2, and each one shaped a design decision reported later in this thesis.

The first concerns the input to the assessment. Published image-based assessments of plant condition commonly require the grower to supply categorical traits alongside the photograph, for example by declaring that the leaves are healthy or that the plant is vigorous. Reported accuracy then depends on the very judgement the system is supposed to replace, and the contribution of the image cannot be separated from the contribution of the declaration. No published system assessed here measures what the photograph alone is worth.

The second concerns validation. Datasets of a few hundred plant images are routinely split at the level of the image. When several images come from the same physical plant, which is the normal case in a shade house where a grower photographs one specimen from several angles, that split places the same plant on both sides, and the reported accuracy measures plant recognition rather than condition assessment. This is a known hazard in the methodological literature, but it is rarely controlled for in applied horticultural work.

The third concerns what a classifier does when shown something outside its training distribution. A model trained only on orchids has no mechanism for declining a question and will assign one of its classes, with high confidence, to a photograph of anything at all. For a system a grower will act on, a confident wrong answer is worse than no answer, and the work surveyed does not treat refusal as part of the task.

1.5 Aim and objectives

The aim of this component is to convert the two breeding decisions from unaided visual judgement into recorded, measured recommendations whose reliability is stated against an explicit baseline. Five objectives follow from that aim. Table 1.1 maps each to the chapter that addresses it and the evidence that settles it.

Table 1.1: Research objectives, the sections that address them and the evidence that settles each

Objective	Addressed in	Evidence
O1 Assess pollination readiness of a Vanda plant from one photograph, without grower-supplied traits	Sections 3.3, 6.1	0.717 accuracy against a 0.639 baseline under grouped cross-validation
O2 Refuse photographs that are not orchids before any verdict is produced	Sections 3.3, 6.2	91.9% of non-orchid images refused, AUC 0.921, 2.2% false refusals
O3 Resolve cross compatibility between two named parents from the registered hybrid record	Sections 3.3, 6.2	Four-tier evidence scheme over 27 registered crosses; no probability reported
O4 Validate every figure against a baseline, on a split that reflects deployment	Sections 3.4, 3.5, 6.1	StratifiedGroupKFold grouped by plant; leakage detected and corrected
O5 Integrate the component without duplicating another component's model	Sections 5.3, 6.4	Component 2's classifier consumed through a declared contract; duplication measured and rejected

1.6 Research questions or hypotheses

RQ1. How much of the pollination readiness decision is recoverable from a photograph alone, measured against the majority-class baseline of the same split?

RQ2. Can a one-class test over a frozen embedding reliably separate orchid photographs from arbitrary images, at a false-refusal rate low enough to be usable in the field?

RQ3. What can honestly be said about a proposed cross when the only available record contains successes and no failures?

RQ4. Does grouping the validation by physical plant change the measured performance of the readiness classifier, and by how much?

The questions are posed rather than framed as hypotheses because three of the four ask how much rather than whether, and the fourth asks what a particular body of evidence will support. None of them has a null form that could be usefully rejected.

1.7 Scope limitations and assumptions

The component is scoped to the genus Vanda and to photographs taken in daylight at roughly arm's length. It ends at the decision to attempt a pollination. The laboratory stages that follow, from capsule harvest through sterile sowing to deflasking, are outside both the data and the scope, and no claim is made about the viability of seed produced by a cross the system has recommended.

Three assumptions are load-bearing. The grower is assumed able to photograph a plant in reasonable light; images taken under coloured artificial lighting fall outside the trained distribution and the input gate will usually refuse them. The labels are assumed to reflect a competent grower's judgement, since no ground truth independent of human assessment exists for readiness. And the registered hybrid record is assumed to be an accurate list of crosses that succeeded, which it is, while explicitly not being assumed to be a list of crosses that are possible, which it is not.

The principal limitation is sample size. Twenty-eight plants is the true sample, not 357 images, and one of the three classes rests on two plants. Every claim about generalisation in this thesis should be read with that number in mind, and Section 7.5 returns to it. A second limitation is coverage at the second level: the knowledge base holds 27 registered crosses, so a grower naming two parents outside it most often receives the no-precedent tier.

1.8 Individual contribution

The following work within project R26-SE-018 is the author's own. The collection and annotation of the 1,190-photograph dataset and of the 543-image gate validation

set. The segmentation and feature-extraction pipeline, and the frozen-embedding representation. The design, fitting and evaluation of the one-class input gate and of the orchid flower filter. The suitability classifier and the four-way model comparison behind it. The grouped evaluation protocol and the identification of the leakage it exposed. The compatibility knowledge base, the name resolution and the evidence tiering. The crossing logbook that records attempted crosses and the stage at which each one stopped. The pollination router and its endpoints. The disease provider interface that consumes Component 2's model. And the four application tabs that present all of it to the grower.

Work contributed to the shared system beyond this component comprises the input-validation pattern now used for optional models across the backend, the capability-based availability check described in Section 5.3, and the secret-handling practice adopted after the repository incident recorded in Section 4.5.

Work that is not the author's and is relied upon is stated here so that the boundary is unambiguous: Component 2's disease classifier, consumed through the provider interface; and the shared Firebase schema and deployment configuration owned by Component 3. Chapter 6 reports the integration between this component and Component 2, and Appendix D reports the measurements that led to consuming that model rather than training another.

1.9 Thesis structure

Chapter 2 reviews the relevant literature and synthesises the gap into a conceptual framework. Chapter 3 sets out the research methodology: the design, the dataset and annotation protocol, the three-stage assessment procedure, the measures and evaluation metrics, the analysis, the ethical position and the research artefacts. Chapter 4 presents the product and solution design, including the requirements, the architecture, the alternatives that were rejected and the detailed design. Chapter 5 describes implementation, integration with the other components, verification and deployment. Chapter 6 reports the results, including the negative ones. Chapter 7 discusses what the results mean, where they are weak and what follows from them.

Chapter 8 addresses commercialisation and sustainability. Chapter 9 concludes and sets out future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review covered four areas: image-based assessment of plant condition, transfer learning with small labelled sets, out-of-distribution detection and open-set recognition, and the horticultural literature on orchid breeding and hybrid registration. Sources were drawn from the ACM and IEEE digital libraries, from agricultural and horticultural journals, and from the published documentation of the Royal Horticultural Society's orchid register. Work was included where it addressed a decision taken from plant imagery, or where it addressed the methodological questions of small-sample validation and of refusal.

One restriction is worth stating. The review did not attempt to cover the general literature on plant disease classification, which is large and which belongs to Component 2 of this project. It is treated here only where this component consumes its output.

2.2 Domain and theoretical foundations

Vanda is a monopodial orchid genus grown across lowland Sri Lanka for cut flowers and as potted specimens. The horticultural requirements for a successful cross are documented and largely uncontested. The pod parent must be mature, vigorous and free of active disease, because it carries the developing capsule for around nine months and the metabolic cost of that is borne entirely by the plant. The flower must be at a receptive stage, typically a few days after opening. The pollen donor's own vigour matters considerably less, since it contributes pollinia and nothing further, and pollinia can be stored under refrigeration.

This asymmetry has a direct consequence for system design that is easy to miss. A photograph-based readiness assessment is worth far more on the pod side than the pollen side, because that is where the cost of a wrong decision falls. The component therefore treats a readiness verdict as advice about which direction to run the cross, not merely as a yes or no about one plant, and Section 3.3 describes the rule that implements this.

The register of orchid hybrids maintained by the Royal Horticultural Society is the authoritative record of which crosses have been made [3]. A grex is registered when a hybrid is first flowered and named, and the register records the seed parent and the pollen parent in that order. Registration is voluntary and carries a fee, which is the structural fact Section 2.4 develops.

On the machine-learning side, the theoretical foundation is transfer learning. A convolutional network trained on a large general corpus learns intermediate representations of shape, texture and colour that remain informative on domains the network never saw [1], [2], [10]. The practical question is how much of the network to retrain, and Section 2.3 reviews the answer for datasets of the size available here.

2.3 Related research

Convolutional networks transfer effectively to plant tasks, and the practical question is usually how much of the network to retrain. Where the labelled set runs to tens of thousands of images, fine-tuning the whole network is normal and produces the best results [1]. Where it runs to a few hundred, retraining the backbone tends to overfit, and using the network as a fixed feature extractor with a light classifier on top is the more robust choice [2].

The residual architecture [2] is a common choice of backbone because its intermediate representations are general and it is small enough to run on a modest server. Hand-engineered descriptors have not been displaced entirely: colour statistics, Gabor texture responses and contour shape remain informative for plant condition, they are interpretable in a way that a learned embedding is not, and they can be computed from a few hundred images without any risk of overfitting a backbone. This component uses them alongside the embedding rather than instead of it.

The methodological literature on small-sample learning is consistent on two points that bear directly on this work. The first is that the effective sample size is the number of independent units, not the number of rows; several photographs of one plant are repeated measurements of a single unit, and treating them as independent

inflates every estimate derived from them. The second is that the validation split must respect that structure, which is what grouped cross-validation provides.

A third body of work addresses what a classifier should do when the input belongs to none of its classes. Approaches divide broadly into methods that calibrate or threshold the classifier's own output, and methods that model the training distribution directly and reject inputs that fall outside it. The second family suits this component better for a reason specific to the data: the suitability classifier has three classes and a few hundred training images, so its output probabilities are not well calibrated and thresholding them would be unreliable, whereas modelling the distribution of the training embeddings does not depend on the classifier at all and can be fitted on all 1,190 photographs rather than only the 357 labelled ones.

Within that family the choice of distance matters. Distance to a class centroid under a Mahalanobis metric is the textbook option, but it assumes the reference set forms a single well-conditioned cloud. Orchid photographs do not: close-up flowers, whole plants and tagged specimens occupy distinct regions of the embedding space, and a centroid falls between them in a region that contains nothing. A k-nearest-neighbour distance makes no such assumption, which is why it was selected here.

2.4 Existing systems products and techniques

Three kinds of system already address parts of this problem, and none of them addresses the decision this component targets.

General-purpose plant identification applications are free, widely installed and technically impressive. They identify species and offer generic care text. They do not assess the condition of a particular plant against a particular decision, and they answer confidently regardless of what they are shown, which is the behaviour Section 1.4 identifies as the third gap. Commercial greenhouse automation addresses irrigation and climate control competently but is priced for industrial glasshouses and has nothing to say about disease or breeding. Agricultural extension services provide exactly the expertise required and are the real substitute for this component, but they are thinly spread and cannot be consulted at the moment a decision is taken.

On the data side, the register of orchid hybrids is the only authoritative record of which crosses have been made, and its structure determines what any system built on it can claim. It records hybrids that were successfully made, flowered and registered. Nobody registers an attempt that produced no capsule. The register therefore has no denominator: it can establish that a pairing is possible, and it can never establish how often such attempts succeed.

This is not a subtle point, but the available secondary sources obscure it. A compilation derived from the register describes intergeneric crossing as moderately to highly successful, while the breeder consulted during this project reports that most intergeneric crosses do not work. Both accounts are honest. The register sees the survivors; the breeder sees the attempts. Any success percentage derived from the register alone would be survivorship bias presented as a measurement, and the second level of this component is designed around that constraint.

2.5 Gap synthesis and conceptual framework

A recurring weakness in the applied papers surveyed is the reporting of a single accuracy figure with no baseline. Where classes are unbalanced, accuracy alone is uninterpretable: a classifier that always predicts the commonest class achieves the majority-class rate while learning nothing. In this component's own data that rate is 0.639, which means a figure in the sixties is not evidence of anything and a figure below it is evidence of harm. Table 2.1 summarises how the reviewed approaches relate to the three gaps of Section 1.4.

Table 2.1: How the reviewed approaches address the three identified gaps, and where this component differs

Approach	Readiness from image alone	Refuses invalid input	Validation respects plant
General plant identification applications	No - identifies species, not condition	No - answers confidently on anything	Not applicable
Fine-tuned CNN on small plant datasets	Partially - often requires declared traits	No - closed-set output	Rarely - image-level splits are common
Frozen backbone with light classifier	Yes in principle	No - the classifier alone cannot refuse	Depends on the protocol used
Open-set and OOD methods	Not a readiness method in itself	Yes - that is their purpose	Not addressed
This component	Yes - traits are measured, not declared	Yes - two-stage gate before any verdict	Yes - grouped by physical plant

The conceptual framework that follows from the table has three commitments, and the whole of Chapter 3 implements them. Refusal precedes classification: no verdict is produced for an input the system cannot place. Measurement precedes declaration: traits are read from the photograph and the grower corrects them only if they wish, rather than supplying them before the system will answer. And the unit of validation is the plant rather than the photograph, because the plant is the independent unit and the photograph is not.

3 RESEARCH METHODOLOGY

3.1 Research design

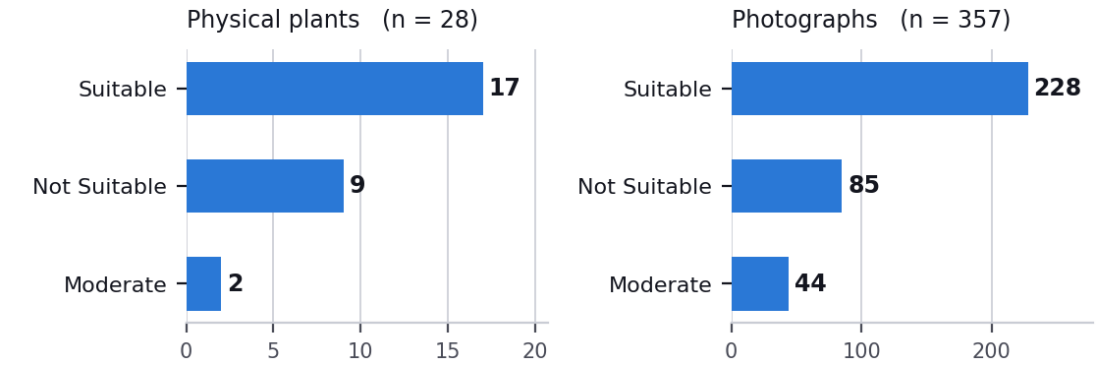
The work follows a design-and-measure cycle rather than a hypothesis test. Each stage of the assessment path was built, measured on data held out from its own fitting, and either retained or replaced on the strength of that measurement. Three parts of the final system exist because an earlier version was measured and found wanting: the input gate exists because the classifier returned a confident verdict for a photograph of a laptop, the second gate stage exists because the first alone admitted other flowers, and the grouped validation protocol exists because the image-level one reported perfect accuracy.

The design is therefore reported here in the order a reader needs it rather than the order in which it happened: the data and how it was labelled, the procedure applied to an image, the measures used to judge the result, the analysis, and the artefacts that make all of it checkable.

3.2 Data participants or study materials

Photographs were taken in working shade houses with a smartphone camera under daylight, at roughly arm's length, without controlled backgrounds or lighting. This was a deliberate choice: a dataset collected under studio conditions would have produced better numbers and a model that fails on the first real photograph it is shown. The growers were informed of the purpose and gave permission; the photographs depict plants rather than people, and no personal data was collected.

The collection totals 1,190 photographs in three groups. 357 are whole-plant images carrying a readiness label and a plant identifier; 300 are close-up flower images; and 533 are tagged specimens photographed with their nursery labels visible, which supply the parent names used at the second level. All 1,190 are used to fit the input gate, because the gate needs to model what an orchid photograph looks like and is not restricted to the labelled subset.



The left panel is the true sample size. Moderate rests on two plants, which is why its per-class scores are reported as indicative only.

Figure 3.1: Composition of the labelled subset by class, shown as physical plants and as photographs. The left panel is the true sample size

Table 3.1 gives the full composition. The difference between the two panels of Figure 3.1 is the single most important fact about this dataset and is referred to repeatedly in what follows.

Table 3.1: Dataset composition. The reference set for the gate is larger than the labelled set because modelling what an orchid looks like does not require a readiness label

Group	Images	Plants or groups	Role
Labelled whole-plant images	357	28 plants	Training and evaluation of the suitability classifier
Close-up flower images	300	part of 375 groups	Gate reference set; flower-condition measurement
Tagged specimens	533	part of 375 groups	Gate reference set; parent-name extraction for level 2
Total project photographs	1,190	375 groups	Reference set for the one-class input gate
Non-orchid validation set	455	11 categories	Held-out gate validation, never used for fitting
Unseen orchid validation set	88	2 categories	Held-out gate validation, web sources
Other-flower negatives	8,089	100 species	Stage 2 training negatives, Oxford Flowers-102 [6]

Each labelled image carries a three-class readiness label and four categorical traits. The traits are recorded for every image so that the contribution of the image can be separated from the contribution of the declaration, which is the first gap identified in Section 1.4; at inference time the system measures them from the photograph instead

of asking for them. Table 3.2 gives the class definitions, and the full written annotation rules are reproduced in Appendix A.

Table 3.2: Readiness classes, their operational definitions and their support in plants and in photographs

Class	Definition	Plants	Images
Suitable	Mature, vigorous, no visible active disease, flowering or in bud; fit to carry a capsule	17	228
Moderate	Usable but compromised; better employed as the pollen donor than as the pod parent	2	44
Not Suitable	Immature, stressed, or showing active disease; should not be bred from this season	9	85

The Moderate class is the weakest part of the annotation and is reported as such throughout. Two plants cannot define a class, and the grouped cross-validation that the rest of this thesis relies on cannot validate it: with two groups, any split places both in training or both in test. Its per-class scores in Chapter 6 are therefore indicative only. The class was retained rather than merged because it carries the direction advice described in Section 3.3.

Labelling was carried out by the author against the written definitions, with ambiguous cases resolved by consulting the grower. No independent second annotator was available, so inter-rater agreement is not reported; this is a threat to construct validity rather than a procedural oversight, and Section 7.5 restates it.

3.3 Experimental or study procedure

An image submitted by the grower passes through three stages, of which the first two can refuse it outright. Figure 3.2 shows the path. This section describes each stage in the order the image meets them, and then the second level of the component, which takes two names rather than an image.

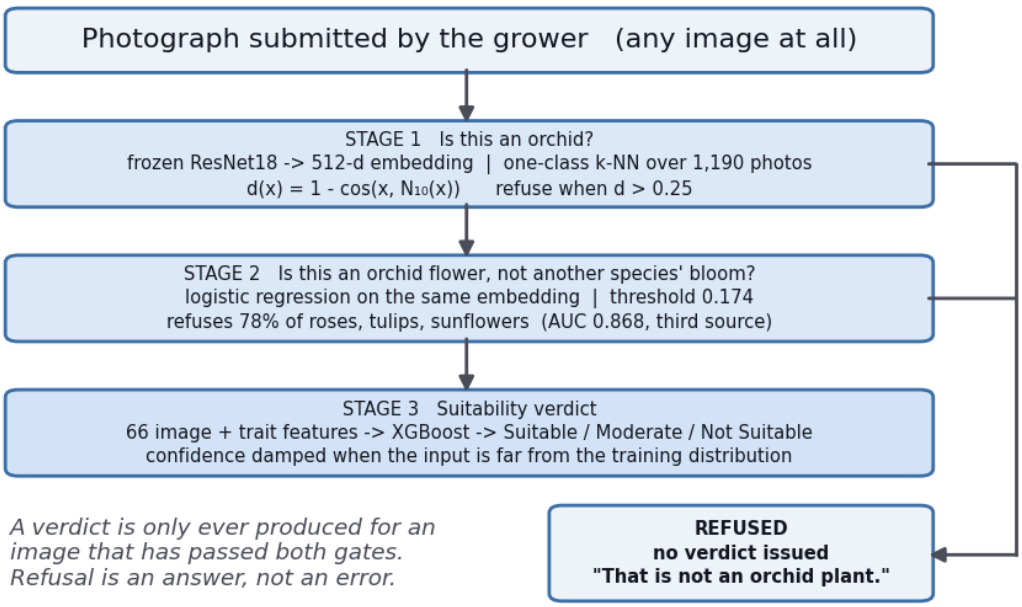


Figure 3.2: The three-stage assessment path. A verdict is produced only for an image that has passed both gates

3.3.1 Pre-processing and segmentation

Images are resized to 224 by 224 pixels for the embedding. For the hand-engineered descriptors the plant is first separated from its background, because a shade-house photograph contains benches, pots, netting and other plants, and colour statistics computed over the whole frame would describe the shade house rather than the specimen. Segmentation uses the Excess Green index, ExG = 2g - r - b over normalised channels [4], to produce an initial vegetation mask, which is then refined by GrabCut [5]. ExG alone is fast and works well on green foliage but is confused by green netting and by brown senescent leaves; GrabCut refines the boundary using the initial mask as a prior.

3.3.2 Feature representation

Two representations are computed and used together. The first is the 512-dimensional output of a ResNet18 pretrained on ImageNet [10] with its final classification layer removed [2]. The backbone is frozen; no gradient reaches it. With 28 independent plants, fine-tuning a network of eleven million parameters would be an exercise in memorising the training set.

The decision to discard the network's 1000-way head rather than use its predictions deserves recording, because the alternative was tried. Asked to classify a photograph of a Vanda flower, the ImageNet head returns sea anemone with high confidence. This is not a failure of the network; ImageNet contains no orchid class and the nearest thing in its label set is a radially symmetric structure with fleshy lobes. The penultimate representation, which encodes shape and texture without committing to a label, is informative. The labels themselves are useless for this domain, and a system built on them would inherit a confident error.

The second representation is 66 hand-engineered features computed over the segmented region, grouped as shown in Figure 3.3 and named in full in Appendix A.

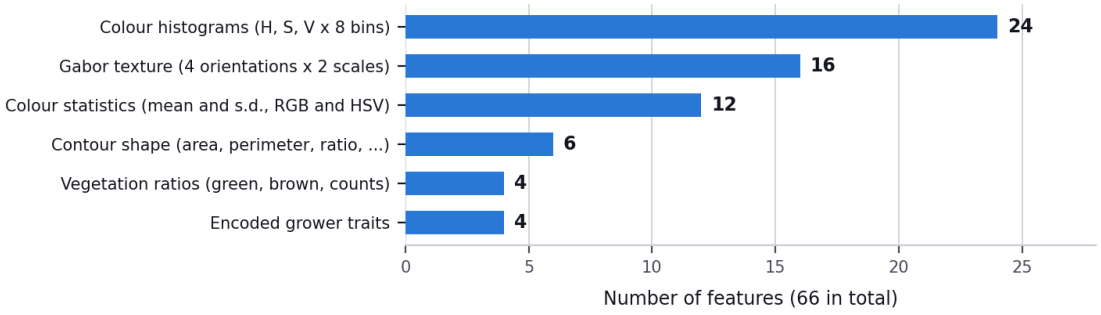


Figure 3.3: Composition of the 66-feature descriptor used by the suitability classifier

The four encoded traits are the mechanism by which the component avoids requiring the grower to declare anything. At training time they are the annotator's values. At inference time each is resolved in order of preference: a value the grower has explicitly corrected in the application is used if present; otherwise the value is measured from the image; otherwise the trait is reported as unresolved and the verdict's confidence is reduced. Appendix C reports how well the two measured traits are actually predicted. Disease is the one trait not measured here; it is obtained from Component 2 as Section 5.3 describes, for reasons Appendix D sets out.

3.3.3 Stage 1: the one-class orchid gate

The gate models the distribution of orchid photographs in embedding space and refuses inputs that fall outside it. All 1,190 project photographs form the reference set. Each embedding is projected onto the unit sphere, so that distance measures

angle rather than magnitude and a dim photograph is not refused for being dim. For a candidate image x the score is the cosine distance to its tenth nearest neighbour in the reference set, d(x) = 1 - cos(x, N10(x)). Using the tenth neighbour rather than the first makes the score robust to a single unusual reference photograph; using a neighbour rather than a centroid makes no assumption that the reference set is unimodal, which Section 2.3 explained it is not.

The threshold is set by quantile rather than chosen by hand. With the keep fraction set to 0.98, the threshold is the value below which 98 per cent of reference distances fall, which on this reference set is 0.25. The median training distance is 0.15 and the ninetieth percentile is 0.21, so the threshold sits well above the bulk of the distribution. Expressing the decision as a keep fraction rather than a raw distance means the operating point retains its meaning if the reference set changes. A secondary rule refuses images whose segmented vegetation fraction falls below two per cent, which catches the few photographs that embed near the reference set for reasons of colour and texture while containing no plant at all.

3.3.4 Stage 2: the orchid flower filter

Stage 1 answers whether an image resembles the reference set. It does not distinguish an orchid bloom from a rose, because both are flowers photographed at similar scale and their embeddings are close. Stage 2 addresses that specifically. A logistic regression is fitted over the same embedding, with the project's 1,190 photographs as positives and 8,089 images from Oxford Flowers-102 [6] spanning 100 species as negatives, together with 80 photographs of non-orchid foliage. The two orchid classes within that dataset are held out of training entirely, so the filter is never taught to call an orchid something other than an orchid. The classifier is balanced by class weight, since negatives outnumber positives roughly seven to one, and the decision threshold is again set by quantile, at 0.174.

Stage 2 runs only on images that passed stage 1. The two are complementary rather than redundant: stage 1 is strong on arbitrary images and weak on other flowers, and stage 2 is the reverse.

3.3.5 Stage 3: suitability classification

An image that passes both gates reaches the classifier. Four candidate algorithms were trained and scored on identical folds: a support vector machine with a radial basis function kernel, a random forest, a multinomial logistic regression and gradient-boosted trees [7]. Reporting all four rather than the winner alone is a deliberate choice, and Section 6.1 explains what it reveals. Their hyperparameters are given in Table 3.3.

Table 3.3: Candidate classifiers and their hyperparameters. A fixed random state of 42 is used throughout

Model	Principal hyperparameters
SVM (RBF)	C = 1.0, gamma = scale, balanced class weight, probability enabled
Random Forest	200 trees, max depth 15, min samples split 5, min samples leaf 2, balanced class weight
Logistic Regression	max iterations 1000, balanced class weight
XGBoost	200 estimators, max depth 8, learning rate 0.1, subsample 0.8, colsample by tree 0.8, mlogloss objective

Feature scaling is fitted inside each fold through a pipeline rather than once over the whole dataset. Fitting the scaler on all the data before splitting would leak information about the test fold into training, which is a milder form of the same error as the image-level split and equally avoidable.

3.3.6 Level 2: cross-compatibility resolution

The second level takes two parent names, the pod parent first by breeding convention, and resolves what the registered record says about that pairing. Resolving the name is the first step and the component reports which of three routes it took, because only one of them is a fact. A name may state its genus, as V. coerulea does; it may state none but appear in the knowledge base, which records one, as a bare grex name such as Kulwadee Maron does; or it may do neither, in which case the collection's genus is used and the result says that it was assumed. The distinction matters because an earlier implementation collapsed all three into a single fallback to Vanda, so any unrecognised name became a Vanda silently and the system returned a confident verdict about a plant the grower had not named. Genera outside the Vanda alliance are held in the vocabulary for the same reason: a cross reaching across an alliance is reported as the botanical impossibility it is, rather than

as an absence of precedent, which reads as a gap in a small knowledge base and invites the attempt. The result is an evidence tier, not a probability; the four tiers are set out in Section 6.2 and the full knowledge base is listed in Appendix E.

Direction is preserved throughout. A crossed with B is a different attempt from B crossed with A, and the component reports a reversed precedent as supporting evidence of a different and weaker kind than an exact one. A nothogenus is expanded into the natural genera it contains before the comparison is made, because a hybrid genus name is itself the record of a cross between those genera: crossing an Aeridovanda back into Aerides introduces no genus the name does not already carry, and treating the nothogenus as atomic made that backcross appear undemonstrated while the same knowledge base held a registered Aeridovanda crossed with Vanda. Where a readiness assessment is supplied for either plant, the rule applied depends on the role. A plant assessed Not Suitable blocks the cross in either role. A plant assessed Moderate does not block it, but the advice differs: in the pod role it suggests reversing the cross, since carrying a capsule for nine months is the heavier demand, while in the pollen role it is accepted without qualification, since a donor needs viable pollinia rather than the vigour to mature a pod. That is the asymmetry established in Section 2.2 made operational.

3.4 Measures and evaluation metrics

Three rules were fixed before results were collected. Every reported figure is accompanied by the majority-class baseline of the same split. Every split reflects the deployment condition rather than the convenience of the data. And a result that falls below its baseline is reported rather than omitted.

For the suitability classifier the reported measures are accuracy, balanced accuracy and macro F1, together with the full confusion matrix. Accuracy alone is not sufficient here because the classes are uneven; balanced accuracy is the harsher measure and the gap between the two is informative about the smallest class. The reference value is the majority-class baseline of 0.639, which is 228 Suitable images out of 357. The equivalent figure at plant level is 17 of 28, or 0.607; the image-level

figure is quoted throughout because it is the baseline of the split on which the classifier is scored.

The gate is measured differently, because it answers a different question: not how accurate the verdict is, but whether a verdict should have been issued at all. Its measures are the proportion of non-orchid images refused, the proportion of genuine orchid photographs falsely refused, and the area under the ROC curve for orchid against non-orchid by distance. These are reported on a validation set assembled from sources never used in fitting, comprising 455 non-orchid images across eleven categories and 88 orchid photographs from web sources the project did not collect.

For the second level no accuracy measure is defined, and this is a finding rather than an omission. The registered record contains no failed attempts, so there is no negative class against which a classifier could be scored. The measure reported is coverage: how many pairings the knowledge base can place in each evidence tier.

3.5 Data analysis

Analysis is by StratifiedGroupKFold cross-validation, which assigns every image of a plant to the same fold while preserving the class distribution across folds as far as the group structure allows. The number of folds is limited by the rarest class: with two Moderate plants, no more than two folds can contain a Moderate group, so the evaluation runs at two folds. Every score reported in Chapter 6 is out of fold, meaning no image was scored by a model that had seen any image of its plant.

The decision to group was not a refinement applied for tidiness. Figure 3.4 shows what it changed.

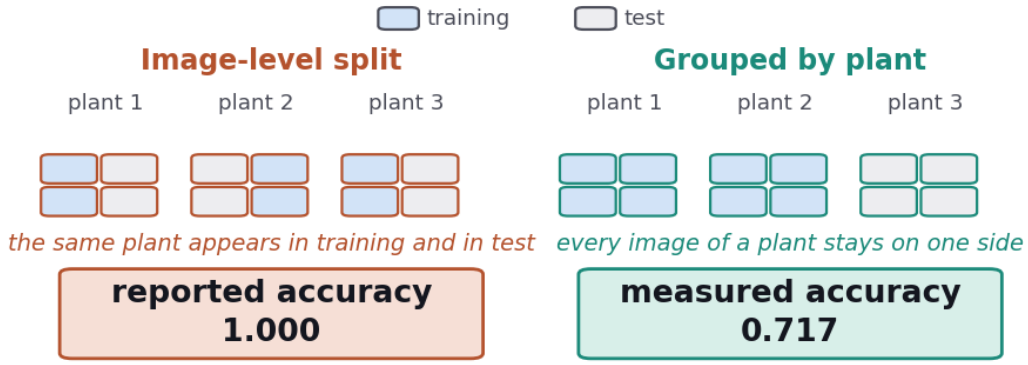


Figure 3.4: The effect of the validation split on the measured accuracy of the same model. Grouping by plant is the difference between measuring condition and measuring identity

An early version of the pipeline reported perfect accuracy on a random image-level split. Because the 357 photographs come from only 28 physical plants, that split placed several images of the same plant on both sides, and the classifier was recognising individual plants rather than their condition. Regrouping the validation by plant reduced the same model, on the same data, to 0.717. The difference of 0.283 is the size of the leakage, and it is reported here because a protocol that detects a defect in its own author's favour is better evidence of its worth than one that merely produces good numbers.

Trait predictors are analysed the same way and are reported in Appendix C. The disease probe reported in Appendix D is analysed leave-one-plant-out rather than by folds, because with 28 plants and a binary outcome the leave-one-out estimate is the less biased of the two.

3.6 Ethics privacy security and responsible AI

Photographs were taken with the growers' permission and depict plants rather than people. The system stores plant records keyed by device and section identifiers rather than personal data, and no grower account exists. The community logbook proposed in Section 9.4 would change that assessment and would require a data protection review before it was built rather than after.

Two responsible-practice decisions are worth stating explicitly. The first is the refusal behaviour: a system a grower acts on should decline a question it has no basis for answering, and the gate exists to make that possible. The second is the decision not to report a success probability for a proposed cross. A plausible number could have been produced from the register and would have been accepted without question by most readers; it would also have been survivorship bias presented as a measurement, and reporting it would have been the more serious failure.

On security, write endpoints require an API key held in a configuration module excluded from version control. This is not a theoretical control: the repository is public and the project's first key was committed and published before the omission was noticed, after which it was rotated and the exclusion added. The incident is recorded rather than quietly fixed because it is the reason secret handling is now treated as a shared requirement across the project.

The registered hybrid record is third-party data, used here in a small attributed research subset. Any commercial deployment would require the rights position to be settled with the publisher rather than assumed. The pretrained backbone [2], [10] and the flower dataset used as stage 2 negatives [6] are used under their respective research licences. Generative AI assistance used in preparing this thesis is disclosed in full in Appendix H.

3.7 Reproducibility and research artefacts

Every figure quoted in Chapter 6 is read from a saved artefact rather than from a laboratory notebook. The training script writes a JSON record containing the evaluation method, the fold count, the grouping variable, the majority baseline, the per-model metrics and confusion matrices, the dataset composition and an explicit list of caveats. The gate and filter scripts write plain-text reports containing their thresholds and their per-category validation results, and the gate additionally writes a row-level CSV of 543 validation images with the distance and the accept decision for each. Appendix G inventories every artefact against the claim it supports.

A fixed random state of 42 is used for every stochastic component. The pinned dependency versions are recorded in the shared requirements file, after a deployment

failure traced to version skew between the environment the models were fitted in and the environment that loaded them. Reproducing any figure in Chapter 6 requires only the dataset, those pins and the training entry point for the stage concerned.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were derived from grower consultation and from the integrated project's shared constraints. Table 4.1 lists them. The three non-functional requirements are as load-bearing as the functional ones: NFR3 in particular is a requirement about what the system must not say, and it rules out an entire class of otherwise attractive designs.

Table 4.1: Functional and non-functional requirements for the component

ID	Requirement	Addressed by
FR1	Assess one plant from a single photograph without requiring declared traits	Sections 3.3.2, 3.3.5
FR2	Refuse an image that is not an orchid, with an explanation rather than an error	Sections 3.3.3, 3.3.4, 4.4
FR3	Allow the grower to correct any measured trait and re-assess	Section 4.4
FR4	Resolve a proposed cross between two named parents and show the evidence	Section 3.3.6
FR5	Rank candidate pollen donors against one pod parent	Section 5.2
FR6	Report the provenance of every trait used in a verdict	Section 3.3.2
NFR1	Return a verdict within five seconds on a mid-range device over mobile data	Section 5.5
NFR2	Fail visibly rather than silently when a model or dependency is unavailable	Section 5.3
NFR3	Never report a success probability for a cross	Sections 3.3.6, 6.2

4.2 Component architecture within the complete solution

Figure 4.1 shows the deployed architecture. The component is a router within the shared FastAPI backend rather than a separate service, which is what guarantees that an image is pre-processed identically regardless of which component received it.

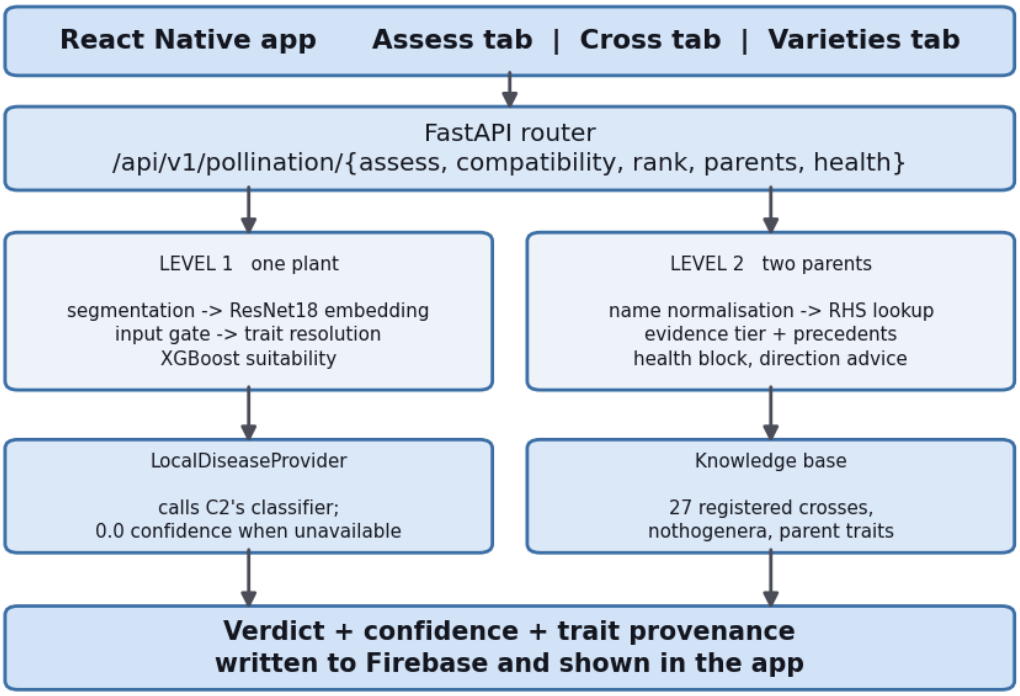


Figure 4.1: Deployed architecture of Component 1, showing the two assessment levels and their dependencies

The two levels are deliberately independent. Level 1 can be used without naming any parent, which is the common case when a grower is surveying a bench; level 2 can be used without a photograph, which is the common case when planning a cross from a catalogue. When both are available, the level 1 verdicts feed the health rules described in Section 3.3.6.

Within the complete solution the component is a consumer as well as a producer. It consumes Component 2's disease classifier through a provider interface, and it writes its verdicts to the shared Firebase subtree owned by Component 3, keyed by the same section identifier the other components use. It produces nothing the others depend on at run time, which is why it was the last component integrated.

4.3 Design alternatives and decisions

Table 4.2 records the seven decisions that most shaped the component, the alternative that was considered in each case, and the reason the decision went the

way it did. Four of the seven were settled by measurement rather than by argument, and the measurements are reported in Chapter 6 and the appendices.

Table 4.2: Principal design decisions, the alternatives considered and the reason each was resolved as it was

Decision	Alternative considered	Reason for the choice
Frozen ResNet18 as feature extractor	Fine-tune the backbone on the labelled set	28 independent plants cannot support fine-tuning eleven million parameters
Discard the ImageNet classification head	Use its predicted labels as features	The head labels a Vanda flower a sea anemone; ImageNet has no orchid class
k-NN distance for the gate	Mahalanobis distance to the class centroid	The reference set is multi-modal; a centroid falls in an empty region between modes
Threshold by quantile (keep fraction)	Fix a raw distance threshold by inspection	A keep fraction retains its meaning if the reference set changes
Two separate gate stages	A single stronger classifier	The two failure modes differ; no single stage strong on both was achieved
Evidence tiers at level 2	A predicted success probability	The register contains no failures and therefore has no denominator
Consume Component 2's disease model	Train a second disease model on this component's images	Measured at 0.522 AUC on whole-plant images, below the majority baseline (Appendix D)

4.4 Detailed design

The component exposes five endpoints under the shared prefix /api/v1/pollination. Table 4.3 documents the three that cross a boundary; the remaining two serve the application's type-ahead. Appendix F gives the full request and response specification for all five.

Table 4.3: Interfaces that cross a component boundary, with the error behaviour each consumer may rely on

Endpoint	Request	Response and error behaviour
POST /assess	Multipart image, optional leaf close-up, and any traits the grower explicitly corrected	Verdict, confidence and trait provenance. Returns HTTP 422 with a structured body when the input gate refuses the image; the application renders this as a result, not a failure
POST /compatibility	Two parent names, pod parent first, plus optional health assessments	Evidence tier and the registered precedents behind it. Never returns a probability
POST /compatibility/rank	One pod parent and a list of candidate donors	Candidates ordered by evidence tier, strongest first

One convention is worth stating because it was the source of a defect. A trait the grower has not answered is omitted from the request entirely; a trait measured and

found absent is sent with its value. An earlier version of the application sent every unanswered field as the string unknown, which the backend could not distinguish from a deliberate answer, and the image-derived value was consequently overwritten every time. The absence of a measurement and a measurement of absence are different facts and are now represented differently.

4.4.1 User interface design

The application presents the component in four tabs. The Assess tab takes a photograph and returns a verdict with its confidence and a card reading what the system read from the photograph, listing each trait beside its origin. Corrections and the optional leaf close-up are collapsed behind disclosure controls rather than presented as forms to fill in, which keeps the default path a single photograph and one tap and satisfies FR1 in the interface as well as in the model.

A refused image produces its own result card explaining that the photograph does not appear to be an orchid, rather than an error dialogue. This is a deliberate interface decision as much as a technical one: a refusal is an answer, and presenting it as a failure would encourage the grower to retry until the system produced a verdict it had no basis for. The Cross tab carries a photograph and a name for each parent in identical cards, drawn by one renderer so that the two sides cannot diverge, and reports on the pair rather than on either half; it will also return the condition half of a verdict when no names have been entered, since a plant in poor condition should not be bred from whatever its tag turns out to say, and only the register half requires a name. The Log tab holds the grower's own crossing records. The Varieties tab lists the parents the knowledge base recognises, which doubles as an honest statement of coverage: a grower can see exactly how much the register knows before relying on it.

4.5 Standards and professional practice

The REST interface follows the shared project convention of versioned prefixes and structured error bodies, so that a consumer can distinguish a refusal from a failure without parsing prose. Python code follows PEP 8 and the project's own requirement that a comment explain why rather than what. Secrets are held in a configuration

module excluded from version control, a practice adopted after the incident described in Section 3.6.

Professionally, the decision with the clearest ethical content is the refusal to report a cross success probability. It would have made a more impressive demonstration and would have been difficult for a non-specialist reader to challenge. The relevant professional obligation is to the grower who would act on that number over a nine-month season, not to the demonstration, and the obligation is discharged by reporting what the evidence supports and naming what it does not.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 lists the technologies used and the role each plays. All are shared with the wider project except the embedding and the classical models, which are specific to this component. The embedding runs under PyTorch [9] and the classical models under scikit-learn [8] with the gradient-boosting implementation of [7].

Table 5.1: Technology stack used by the component

Layer	Technology	Role in this component
Application	React Native with Expo	Assess, Cross and Varieties tabs; camera and gallery access
Backend	FastAPI on Python	Pollination router, model registry, shared pre-processing
Embedding	PyTorch, ResNet18 pretrained on ImageNet	512-dimensional frozen feature extractor
Classical models	scikit-learn, XGBoost	Suitability classifier, flower filter, one-class gate, trait predictors
Imaging	OpenCV, NumPy	Segmentation, colour, texture and shape descriptors
Shared state	Firebase Realtime Database	Verdicts written per plant and visible to the other components
Serialisation	joblib	Model persistence, with the unwrapping rule of Section 5.2

5.2 Component implementation

The component is implemented across a set of focused modules rather than a single pipeline script. Segmentation, feature extraction, the embedding, the gate, the flower filter, trait resolution, the suitability predictor and the compatibility knowledge base are separate, each with its own training or fitting entry point and its own saved artefact. The separation is what makes the staged evaluation in Chapter 6 possible: each stage can be measured without running the others.

The crossing logbook is implemented as a separate service with its own store, because what it holds is not a model artefact but the grower's record. A cross moves through a pipeline of stages and leaves it at the gate where it stopped, and the gates are kept apart deliberately: a flower that dropped without setting is usually technique, timing or non-viable pollen; a capsule aborted during development is usually the pod parent's condition; a capsule that matured but held no viable seed is where genetic

incompatibility actually appears. A single success-or-failure flag would collapse the three and discard the only signal the record exists to collect. A record whose outcome is unknown is kept in a fourth state that is counted separately from the failures, since abandoned records that quietly became failures would bias every figure the store reports. Writes go to a local file first and to the shared database second, in that order and never the reverse, because the grower is standing in a shade house and the record must be written whether or not there is a signal; the remote copy is what makes it survive the device.

Trait resolution deserves a note because it carries the design commitment of Section 3.3.2. It takes the grower's explicit corrections, the image-derived measurements and the disease provider's output, and returns both a resolved value and its origin for each trait. An earlier version had a pending state for traits that were neither supplied nor measurable, which proved to be a source of confusion rather than information and was removed in favour of reporting the trait as unresolved and damping the confidence.

Flower presence is measured from the segmented region using a saturation and area rule rather than a learned model, for reasons Section 6.2 sets out. The thresholds are tuned for precision rather than recall: the system would rather miss a washed-out bloom than report a flower on a seedling, because the second error leads a grower to attempt a pollination that cannot succeed.

A practical note on serialisation is recorded here because it cost a deployment. The training script wraps the gradient-boosting classifier in a helper class that remaps non-contiguous class labels, since the library version pinned in production requires labels beginning at zero. The final model is unwrapped before it is serialised, so that the saved file does not depend on a class defined in a training script that is not deployed to the server.

5.3 Integration with the other components

The component integrates with Component 2 through a provider interface. The provider exposes a single availability check and a single classification call, and returns a condition with a severity on a three-point scale. The interface is a declared

contract rather than a direct import, so that Component 2's internals can change without this component being rebuilt. Section 6.4 reports how that integration behaves in practice, and Appendix D reports the measurements that led to it.

The availability check is the part worth recording. An earlier version reported itself available whenever the model file existed on disk. On a server without the deep-learning runtime installed, the file existed, the check passed, the import then failed inside the call, and the system fell back to a colour heuristic while continuing to report that the trained classifier was in use. The corrected check verifies that the runtime can actually be imported. That pattern, checking the capability rather than the artefact, is now used for every optional dependency in the backend and is one of this component's contributions to the shared system.

The failure semantics are equally deliberate. If Component 2 is unavailable the provider returns zero confidence and this component proceeds on image evidence alone. It does not treat an unidentifiable leaf as a healthy one. Integration with Component 3 is one-way and passive: verdicts are written to the shared subtree using the schema that component owns, and nothing in this component reads from it.

5.4 Verification and software testing

Table 5.2 lists the behaviours that were verified and what each check establishes. These are the behaviours a consumer of the component is entitled to rely on, so they are stated as contracts rather than as implementation details.

Table 5.2: Verification categories for the component

Category	What is verified
Gate behaviour	A non-orchid image produces a structured 422 carrying the gate's own measurements, not a generic failure
Trait precedence	An explicitly corrected trait overrides the measured value; an omitted trait does not overwrite it
Disease provider	An unavailable model yields confidence 0.0 and is not collapsed into a report of no disease
Direction handling	A x B and B x A resolve to different results; a reversed precedent is reported as weaker evidence
Health rules	A Not Suitable parent blocks the cross; a Moderate parent produces direction advice without blocking
Serialisation	Every saved model loads in a clean environment matching the pinned production versions
End-to-end	Photograph to recorded verdict against a running backend, including the refusal path

These checks were carried out by hand against a running backend rather than by an automated regression suite, and are reported that way rather than as a pass rate. The component's verification therefore rests on two things: those documented interface checks, and the measured validation artefacts that every figure in Chapter 6 is read from. Appendix G lists both, artefact by artefact, with the claim each one supports. Converting the interface checks into automated tests is named as outstanding work in Section 9.4, and it is the one piece of engineering rigour this component is missing rather than merely short of.

5.5 Deployment and operation

The backend is deployed to a virtual private server reachable over a domain name, with the application pointing at it through a single configuration module. Models are loaded at process start and held in memory; the health endpoint reports which are loaded, so a partially provisioned server is visible rather than silent. The assessment path is budgeted at under five seconds end to end on a mid-range Android device over mobile data, which NFR1 requires and which the single forward pass through a frozen backbone comfortably allows.

The operational lesson of this component concerns environment parity rather than deployment mechanics. The first deployment to a shared server failed not because the code was wrong but because the environment differed from the one the models had been fitted in: the numerical library had changed its internal module layout between the version that wrote the files and the version that read them, and a gradient-boosting library present in development was absent in production. The remedy, refitting the serialised models inside an environment that matches the server's pinned versions and verifying that each loads before release, is now a release step rather than a debugging technique.

6 RESULTS AND VALIDATION

6.1 Research results

The research question of Section 1.6 asks how much of the readiness decision is recoverable from a photograph alone. Four candidate classifiers were scored on identical grouped folds to answer it. Figure 6.1 reports accuracy and balanced accuracy side by side, and Table 6.1 gives the full numbers.

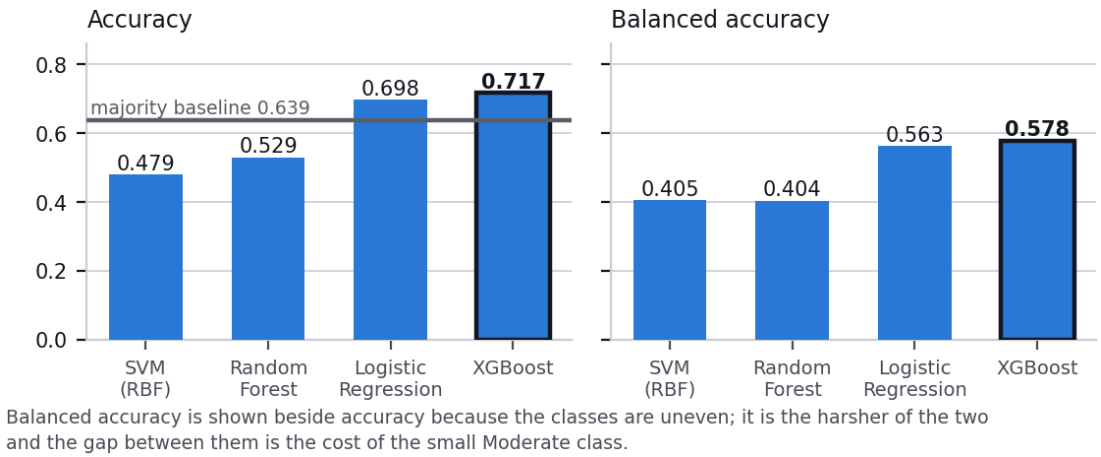


Figure 6.1: Model selection under StratifiedGroupKFold grouped by plant. Two of the four candidates fall below the majority-class baseline

Table 6.1: Candidate classifiers scored out of fold on identical grouped splits, against a majority-class baseline of 0.639

Model	Accuracy	Balanced accuracy	Macro F1	Beats baseline
SVM (RBF)	0.479	0.405	0.479	No
Random Forest	0.529	0.404	0.525	No
Logistic Regression	0.698	0.563	0.695	Yes
XGBoost (deployed)	0.717	0.578	0.715	Yes

Two of the four candidates perform worse than always answering Suitable. That is a more useful result than it first appears. It establishes that the margin achieved by the selected model is a property of that model rather than of the task being easy, and it is exactly the comparison a report quoting only the winning figure would conceal. It also shows how little headroom the task offers: the gap between the baseline and a perfect classifier is 0.361, and the deployed model captures 0.078 of it.

Balanced accuracy is reported beside accuracy because the classes are uneven. It is the harsher measure, and the gap between the two columns is largely the cost of the two-plant Moderate class, which the confusion matrix in Figure 6.2 exposes directly.

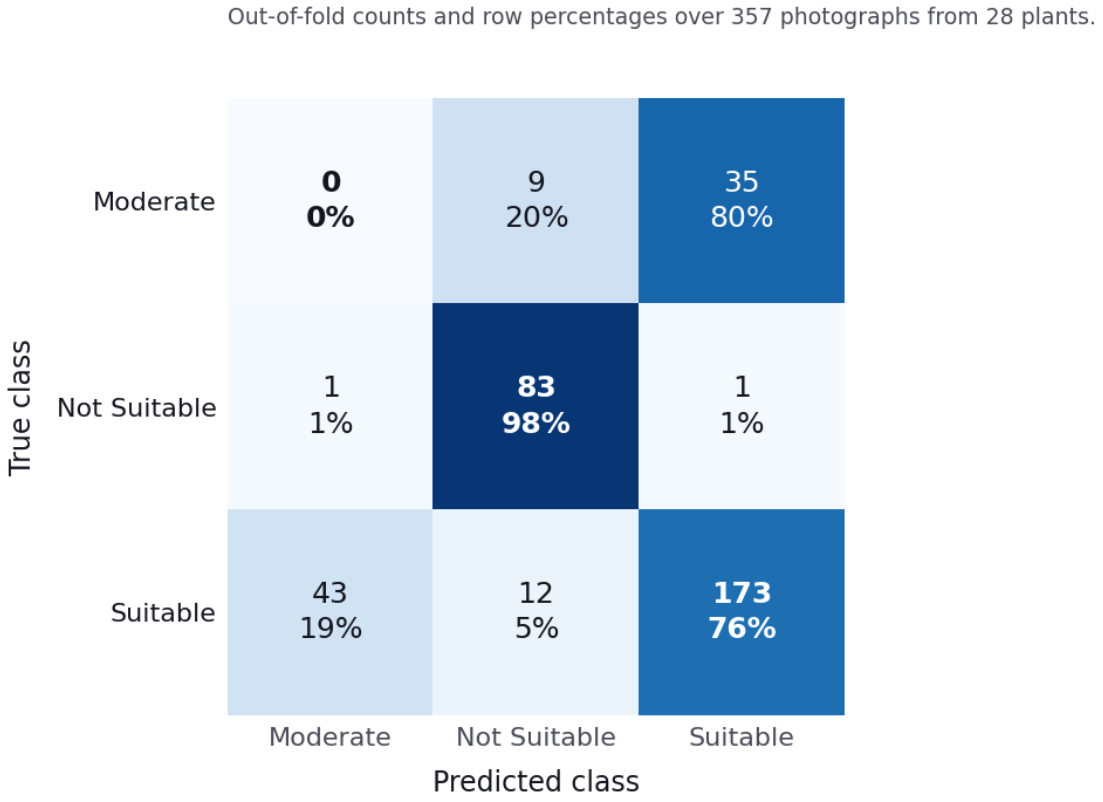


Figure 6.2: Out-of-fold confusion matrix for the deployed classifier, with counts and row percentages

Table 6.2: Per-class performance of the deployed classifier

True class	n	Recalled correctly	Principal confusion
Not Suitable	85	83 (98%)	One image each to Moderate and Suitable
Suitable	228	173 (76%)	43 to Moderate, 12 to Not Suitable
Moderate	44	0 (0%)	35 to Suitable, 9 to Not Suitable

Table 6.2 summarises the per-class behaviour. The matrix exposes the component's real limitation and also its most useful property. Not Suitable is recovered at 98 per cent, which is the direction that matters most: the system rarely tells a grower to breed from a plant that should not be bred from, and that error is the expensive one.

Suitable is recovered at 76 per cent, with most of the loss going to Moderate, which in practice means advice to use the plant as a pollen donor rather than a refusal.

Moderate is never predicted correctly. With two plants behind the class, the grouped cross-validation can neither learn it nor validate it, and its scores are indicative only. It is worth being precise about what this does and does not mean. It does not mean the class is wrong; it means the dataset cannot yet support it. Enlarging it is the first item of future work in Section 9.4.

RQ4 is answered by Section 3.5 and is restated here because it is a research result in its own right: the same model on the same data scored 1.000 under an image-level split and 0.717 under a split grouped by plant. The 0.283 difference is the leakage, and it is the clearest evidence this thesis offers that the validation protocol, not the model, determines what an accuracy figure means.

6.2 Product and technical results

The gate and the second level are reported here rather than in Section 6.1 because they are properties of the delivered product rather than answers to a research question about prediction accuracy. Both nevertheless rest on measurement.

6.2.1 Stage 1 input gate

The gate was evaluated in two ways. Grouped five-fold cross-validation over the 1,190 project photographs measures how often it refuses a genuine orchid photograph of the kind it was fitted on: 2.2 per cent, against a threshold set to keep 98 per cent of training distances, which is the expected agreement between the operating point and its outcome. The held-out validation set measures what matters more, namely behaviour on images from sources the gate has never seen. Figure 6.3 and Table 6.3 report it.

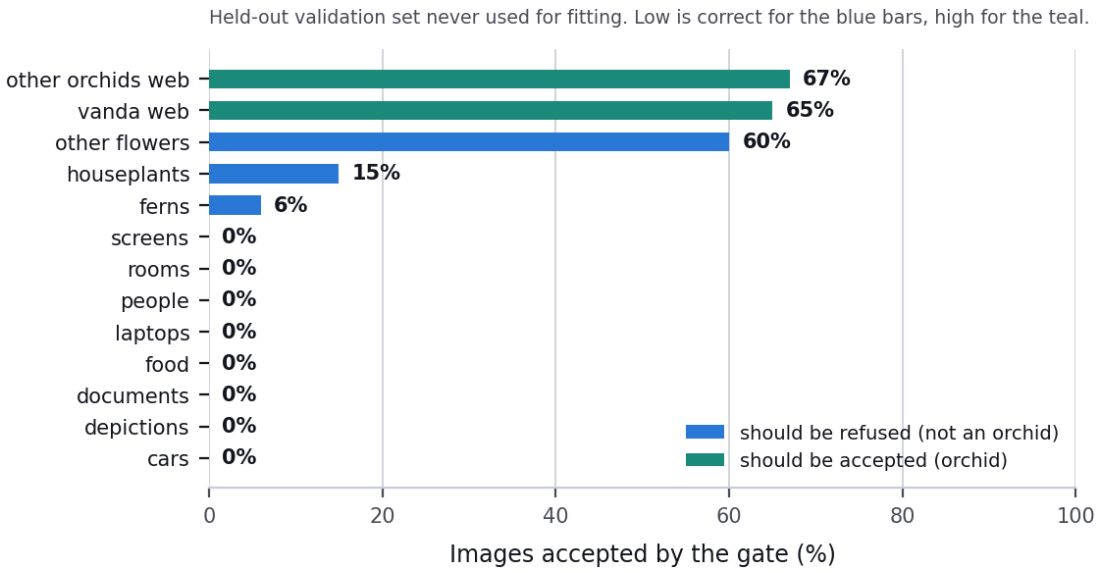


Figure 6.3: Gate acceptance by category on the held-out validation set. Low is correct for the blue bars and high for the teal

Table 6.3: Stage 1 gate measured on 455 non-orchid and 88 unseen orchid photographs, none used in fitting

Category	n	Accepted	Median distance	Reading
Cars, documents, food, laptops, people, rooms, screens	309	0%	0.4	No non-plant image was accepted
Depictions of orchids	2	0%	0.3	A magnet and a lithograph, both refused
Ferns	51	6%	0.3	Refused despite similar leaf geometry
Houseplants	48	15%	0.3	Foliage is the hardest negative class
Other flowers	45	60%	0.2	Passes stage 1; stage 2 exists for this
Vanda, unseen web photographs	43	65%	0.2	Accepted; the remainder are false refusals
Other orchid genera, web	45	67%	0.2	Accepted as orchids

Overall the gate refuses 418 of 455 non-orchid images, or 91.9 per cent, at an area under the ROC curve of 0.921 for orchid against non-orchid by distance. The distribution of errors is as important as the rate. Nothing that is not a plant is ever accepted. Foliage is harder, and other flowers are hardest, which is precisely the division of labour stage 2 was built for.

The cost of this conservatism is that 34 per cent of genuine orchid photographs from unfamiliar web sources are refused. For this application that trade is the right way round: a refusal costs the grower a retake, whereas a confident verdict on the wrong species costs a breeding season. It is also worth noting that the web photographs are not the deployment condition; the 2.2 per cent figure is the one that describes photographs taken the way the system expects.

6.2.2 Stage 2 flower filter, and a negative result retained

The filter was fitted on 1,190 positives against 8,584 negatives at a threshold of 0.174. On a third-party test never used in training it accepts 82 per cent of orchid photographs and refuses 78 per cent of roses, tulips and sunflowers, at an area under the curve of 0.868. Combined with stage 1, the share of other flowers reaching the classifier falls substantially from the 60 per cent that pass stage 1 alone.

A separate line of work on flower detection produced the clearest negative result in this component, and it is retained in the record rather than omitted. A model trained to detect the presence of a bloom from clean cut-out images reached 0.999 cross-validated accuracy. Tested on real nursery photographs it classified 11 of 11 flowering plants correctly and 0 of 38 non-flowering plants correctly, as Figure 6.4 shows.

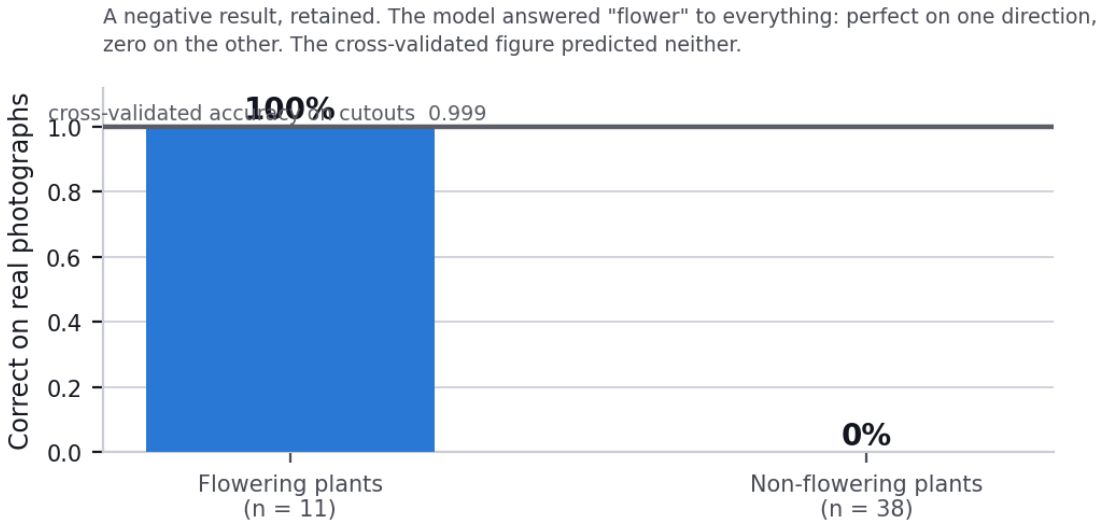


Figure 6.4: A model that answered "flower" to everything. Cross-validated accuracy on cut-outs predicted neither direction of the real-photograph result

The model had learned to answer flower to everything. On a test set containing only flowering plants that behaviour scores perfectly, which is why testing both directions matters. This is the sharpest available evidence in this project that cross-validated accuracy on a convenient dataset predicts very little about deployment, and it is the reason flower presence is measured in the deployed system by an explicit saturation and area rule whose failure modes are visible, rather than by a model whose confidence is unearned.

6.2.3 Cross-compatibility coverage

The knowledge base holds 27 registered crosses drawn from the orchid register [3], listed in full in Appendix E. Fourteen are interspecific and thirteen intergeneric. Two defects in the resolution layer were found by inspection against this same knowledge base and are reported because both produced confident wrong answers rather than visible failures. An unrecognised parent name was read as a Vanda without saying so, so a misspelt or foreign genus returned a verdict about a different plant; and a nothogenus was compared as an atomic genus, so crossing an Aeridovanda back into Aerides reported no precedent while the register extract in use held an Aeridovanda crossed with Vanda. Both are corrected as described in Section 3.3.6, and the correction was checked against eight pairings spanning all four tiers and the cross-alliance case, with no change to the tier assigned to any previously correct pairing. Table 6.4 sets out the four evidence tiers and what each permits the system to say.

Table 6.4: The four evidence tiers and the claim each supports

Tier	Condition	What the system reports
Registered precedent	This exact pairing, in this direction, appears in the register	The precedents themselves, with grex name and year
Genus combination proven	This genus pairing is registered, though not this exact cross	That the genera are known to cross, with examples
No registered precedent	Neither the pairing nor the genus combination is registered	That no precedent was found, and explicitly not that the cross will fail
Not advised	A supplied readiness assessment rules one parent out	Which parent, and why, with direction advice where applicable

The third tier carries the weight of Section 2.4 and its wording was chosen with care. Absence from the register means one of four things, and the system cannot distinguish them: nobody has tried the cross, someone tried and it failed, someone

succeeded and did not register, or someone succeeded and the offspring were not worth naming. Reporting absence as evidence of failure would assert the second of those four without grounds.

Coverage is the obvious limitation. Twenty-seven crosses is a small subset of the register, and a grower naming two parents outside it receives the third tier most of the time. The honest framing is that the second level is a precedent lookup whose value scales with the knowledge base, not a predictive model whose value scales with training.

6.3 User or stakeholder validation

Validation by stakeholders took two forms, and both are reported with their scale stated, because a small number of participants is a real limit on what can be concluded.

Growers were shown the application and asked to carry out representative tasks: photograph a plant on their own bench, read the verdict, and say whether they agreed with it. Two findings came out of that and both changed the product. The first was that the original interface asked for the plant's condition before it would return a verdict, and growers reasonably asked why they were being asked to supply the answer. That observation is the origin of the measured-traits design of Section 3.3.2 and of requirement FR1. The second was that an early build presented a refused image as a network error, and growers responded by photographing the same plant repeatedly until something came back. The refusal card described in Section 4.4.1 was built in response.

The breeder consulted on crossing reviewed the second level's output and confirmed the decision not to report a success percentage, on the grounds that the registered record does not contain the failed attempts such a figure would require. That confirmation is the reason the evidence tiering survived into the delivered product rather than being replaced by a more impressive-looking number.

What this validation does not establish should be stated plainly. The number of participants was small, the sessions were not structured as a formal usability study, no task-completion times or error rates were recorded, and no satisfaction instrument

was administered. The findings above are therefore design evidence, not measured usability. A structured study with a defined participant count and a validated instrument remains outstanding and is named in Section 9.4.

6.4 Integration results

Integration with Component 2 was the one run-time dependency this component accepted, and it is reported here because the decision to accept it was itself a measured one. The alternative, training a disease model on this component's own whole-plant photographs, was attempted first and abandoned on evidence.

A disease signal was probed directly from the 357 whole-plant images. The strongest individual descriptor reached an area under the curve of 0.700, and a classifier combining them reached 0.522 overall, validated leave-one-plant-out. The accuracy of 0.571 is below the 0.643 obtained by answering healthy for every plant. A model that is worse than the majority class is not a weak model; it is evidence that the signal is not present in this input. Appendix D reports the full measurements.

The reason is not mysterious. A lesion occupies a few square centimetres of one leaf, and a whole-plant photograph taken at arm's length renders that as a handful of pixels. Component 2 photographs a single leaf at close range on 28,782 training images, which is the right input and roughly eighty times the data. Duplicating it here would have produced a second and worse answer to a question already answered, and the two would have drifted apart as soon as either was retrained.

The integration itself behaves as specified. Component 2's classifier is called through the provider interface and its severity is folded into this component's trait vector on a three-point scale. When the provider is unavailable it returns a confidence of zero and this component proceeds on image evidence alone, rather than treating an unidentifiable leaf as a healthy one. The fault that made this worth testing is recorded in Section 5.3: an availability check that inspected the model file rather than the runtime reported success on a server where the import would fail, and the system silently used a colour heuristic while claiming to use the trained model.

Integration with Component 3 is one-way: verdicts are written to the shared Firebase subtree under the section identifier that component owns. No schema change was

required in either direction, which is the practical benefit of having agreed the database shape before any model was trained.

6.5 Summary of findings

RQ1 is answered: 0.717 accuracy and 0.715 macro F1 against a 0.639 baseline, measured out of fold on a split grouped by plant, with no grower-declared traits at inference time. The margin is real but modest, and the class distribution explains why it cannot be large.

RQ2 is answered: the gate refuses 91.9 per cent of non-orchid images at an area under the curve of 0.921, refuses every one of the 309 images in the seven non-plant categories, and falsely refuses 2.2 per cent of photographs of the kind it was fitted on.

RQ3 is answered in the negative, which is itself the finding: nothing can honestly be said about the success rate of a proposed cross from a register that contains no failures, and the component reports evidence tiers instead.

RQ4 is answered: grouping the validation by plant changed the measured accuracy of the same model from 1.000 to 0.717. The difference is not a refinement; it is the difference between measuring plant identity and measuring plant condition.

Two results outside the research questions are worth carrying forward. The cut-out flower model reached 0.999 cross-validated accuracy and 0 of 38 on real non-flowering photographs, and disease prediction from whole-plant images fell below its own majority baseline. Both are negative, both are retained, and both changed the delivered system.

7 DISCUSSION

7.1 Interpretation of findings

The headline number deserves to be read carefully. An accuracy of 0.717 against a baseline of 0.639 is a margin of 0.078, which in isolation sounds slight. Two things qualify it. The first is that the available headroom is 0.361, so the model captures roughly a fifth of what there is to capture; that is a modest result but not a trivial one on 28 plants. The second is that the margin is unevenly distributed across the classes in a way that favours the application: the class the system recovers almost perfectly is the one whose misclassification is expensive.

The gate results are stronger and, in a sense, more interesting. A one-class test over a frozen embedding, fitted without a single negative example, refuses every car, document, laptop, room, screen, meal and person it was shown. That it does so without having been told what any of those are is the property that makes the approach practical: the set of things a grower might photograph by accident cannot be enumerated in advance, so a method that requires negative examples of each would not survive contact with a shade house.

The flower-detection failure is the result this author would keep if only one could be retained. A model with 0.999 cross-validated accuracy that is correct on 0 of 38 real photographs in one direction is a complete and compact demonstration that the validation set defines what the number means. The same lesson, arrived at independently, is what the leakage result of Section 3.5 teaches about splits.

7.2 Comparison with literature baselines and existing solutions

Direct numerical comparison with published work is not possible and claiming otherwise would be misleading. No published dataset of Vanda readiness exists, the class definitions here are specific to this project, and an accuracy measured on 28 plants is not commensurable with one measured on a public benchmark. The comparison that can be made is methodological, and it is made against the baseline that matters most, which is the majority class of this component's own split.

Against the general plant-identification applications reviewed in Section 2.4, this component differs in answering a condition question rather than an identity one, and in declining to answer at all when shown something outside its domain. Against the small-sample transfer-learning literature it follows the recommended practice of freezing the backbone, and departs from common applied practice in grouping the validation by physical unit. Against the open-set literature it uses a standard distance-based rejection method, and the contribution is not the method but the demonstration that it is fittable on 1,190 unlabelled in-domain photographs and effective on categories it never saw.

Against extension services, which are the real substitute, the comparison is not one of accuracy at all. An extension officer is more accurate than this component on every question it addresses. The difference is availability at the moment the decision is taken, and that is the claim the component should be judged on.

7.3 Research contribution

The research contribution of this component is fourfold. First, it establishes a measured figure for how much of the Vanda readiness judgement is recoverable from a photograph with no declared traits, on a protocol that does not flatter itself. Second, it demonstrates that a one-class gate fitted on in-domain photographs alone generalises to arbitrary out-of-domain categories, which matters for any applied system whose users can photograph anything.

Third, it documents a leakage mechanism and its magnitude on a real dataset: the same model, the same data, 1.000 under an image-level split and 0.717 under a grouped one. Published warnings about this hazard are usually general; a measured instance with the size of the error attached is more useful to a practitioner deciding whether the warning applies to them.

Fourth, and negatively, it establishes that the orchid register cannot support a cross success probability, and that the correct response is an evidence tier rather than a number. Stating the limit of what the data can bear is part of answering the research question, not a failure to answer it.

7.4 Product contribution

As a product the component gives a grower two things that did not previously exist in usable form. The first is a recorded readiness assessment from a photograph taken on a phone already in their pocket, with the reasoning visible: each trait is shown beside its origin, and the confidence is damped rather than hidden when the evidence is thin. The second is a precedent lookup for a proposed cross that does not pretend to more than it knows.

The input gate is a product contribution in its own right and has been adopted as the architectural pattern for optional models elsewhere in the integrated system. So has the capability-based availability check of Section 5.3. The general lesson behind both, that a component should fail visibly rather than silently and should decline a question it cannot answer, applies to all four components of the project.

7.5 Limitations and threats to validity

The most serious limitation is sample size. Twenty-eight plants is the true sample, and no amount of further photography of the same plants will change it. The Moderate class, with two plants, cannot be validated at all. Every generalisation claim in this thesis is bounded by that number, and the two-fold cross-validation it forces is coarser than would be chosen freely.

The second is annotation, and it is a threat to construct validity rather than a procedural oversight. Labels were assigned by a single annotator against written definitions, with the grower consulted on ambiguous cases, but no second independent annotator was available and inter-rater agreement is therefore unmeasured. Since readiness has no ground truth independent of expert judgement, the honest statement is that the model predicts this annotator's judgement as informed by this grower, and the degree to which that generalises to other expert judgement is unknown.

The third is domain coverage, a threat to external validity. All photographs come from a small number of shade houses in one country under daylight. The gate limits the damage by refusing out-of-distribution inputs rather than guessing at them, and the confidence damping reduces the strength of verdicts near the boundary, but

neither mechanism converts an out-of-domain photograph into a reliable one. The 34 per cent false-refusal rate on unfamiliar web photographs is a direct measurement of how narrow the trained domain is.

The fourth is coverage at the second level: 27 crosses. The mechanism is sound and the tiers are honest; the knowledge base is small. The fifth is that the photograph and the name are independent pieces of evidence that the system cannot reconcile. No model in this component identifies a variety from an image, so nothing verifies that the plant photographed is the plant named, and a grower who photographs one plant while typing another's tag receives a confident verdict about a plant that does not exist. The interface states this rather than concealing it, and the logbook removes it for records created there by capturing the photograph and the name into one plant record, but it remains true of the Cross tab and cannot be closed without variety identification, which would require a labelled multi-variety dataset this project does not have. The sixth is test coverage. The repository does run an automated suite, and it is not decorative: it rejected a change to this component that reached the shared database outside the agreed chokepoint, which no behavioural test would have noticed. But that suite protects project-wide invariants, not the interface contracts of Table 5.2, which remain verified once rather than continuously.

7.6 Implications

For the grower, the implication is narrower than the headline figure suggests and more useful than it sounds. The component is not accurate enough to replace judgement, and it should not be presented as doing so. It is accurate enough, in the one direction that matters, to act as a check: a plant it calls Not Suitable is very likely to be unfit, and that is the error whose cost is a season. The right framing for deployment is advisory, and the interface was built to reinforce that by showing the evidence behind every verdict.

For the project, the implication is that refusal belongs in the architecture rather than in any one model. The gate began as a fix for a single embarrassing behaviour and became the pattern by which the whole backend handles optional models and

unfamiliar inputs. A system that four students build in parallel will fail at its seams, and the seams are exactly where silent fallbacks hide.

For practitioners applying machine learning to small agricultural datasets, two implications follow directly from the measurements. The unit of validation must be the physical specimen, not the photograph, and the cost of getting this wrong is measured here at 0.283 accuracy. And cross-validated accuracy on a convenient dataset should be treated as a lower bound on optimism rather than an estimate of deployment performance; the cut-out flower model's 0.999 is the demonstration.

For the research question of cross compatibility, the implication is that progress requires different data rather than a better model. No amount of modelling will extract a failure rate from a record of successes. Section 9.4 sets out what collecting that data would involve.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value offered by this component is the conversion of a nine-month gamble into a recorded, measured decision taken before the gamble is made. The primary user is the smallholder grower who breeds a handful of crosses a season and cannot afford to lose one to a plant that was never fit. The secondary and commercially more significant user is the nursery running a breeding programme, for which the compatibility record and the crossing logbook together are the more valuable half, because they turn knowledge that currently leaves with the person who holds it into an institutional record. The botanical garden consulted for this project keeps its crossing records in handwritten books and reported losing part of them to flooding; a record that is legible, searchable and held off the premises is a concrete benefit that requires no model to be correct.

The user least often named is the new entrant. The barrier to breeding Vanda commercially is not capital; the plants and the shade house are modest investments by agricultural standards. It is the several seasons of losses that teach a grower to read a plant. A system that shortens that period lowers a real barrier to a high-value crop, and that is the mechanism behind the Goal 8 claim in Section 8.5.

8.2 Market and competitive position

Three alternatives exist, and Section 2.4 reviewed them as techniques; here they are considered as competitors. General-purpose plant identification applications are free and widely installed, but they identify species and offer generic care text, and they answer confidently regardless of what they are shown. Commercial greenhouse automation is priced for industrial glasshouses and has nothing to say about breeding. Agricultural extension services provide exactly the expertise required and are the real substitute, but they are thinly spread and cannot be consulted at the moment the decision is taken.

The differentiation claimed is therefore not accuracy in the abstract. It is that the component answers a breeding question at smallholder scale, from a photograph, and

declines to answer when the evidence does not support an answer. The refusal behaviour is a differentiator rather than a limitation: it is precisely the property that separates this system from an identification application that will confidently classify a photograph of a desk.

8.3 Business or sustainability model

The model that fits the user described above is a free application with optional paid hardware elsewhere in the integrated system. Photograph-based assessment requires no sensor node and costs only server capacity, which matters because the smallholders who would benefit most are the least able to pay in advance for software whose value they have not yet seen. The marginal cost of an additional user of this component is one forward pass through a frozen ResNet18 and one gradient-boosted prediction, which is small and constant.

The realistic distribution channel is the agricultural extension service and the orchid growers' associations rather than an app store listing. Both already hold the trust this component has to borrow, and both reach smallholders who do not go looking for agricultural software. A nursery partnership would serve the second user group, since a nursery running a breeding programme has both the volume to make a compatibility record worth maintaining and the staff to maintain it.

Two statements are owed plainly. No revenue has been realised: this is an academic prototype and has never been sold. And no pricing has been tested with growers, so the free-application model above is an argument from the users' circumstances rather than a validated business decision.

8.4 Scalability intellectual property and compliance

The component scales straightforwardly in users, since the models are fixed at inference and the per-request cost is small and constant. It does not scale straightforwardly in accuracy: more users do not improve the model unless their photographs are labelled, and labelling requires expert judgement. The growth path that does improve it is the grower logbook proposed in Section 9.4, which collects outcomes rather than opinions and therefore needs no annotator.

On intellectual property, the registered hybrid record is third-party data used here as a small attributed research subset; commercial use would require the rights position to be settled with the publisher rather than assumed. Photographs of growers' plants were collected with permission for this research, and a commercial deployment would need that permission re-obtained on commercial terms. The pretrained backbone [2], [10] and the flower dataset used as stage 2 negatives [6] are used under their respective research licences.

On compliance, the current system stores plant records rather than personal data, and the identifiers are device and section keys rather than people. Adding grower accounts, which the community logbook would require, changes that assessment and would require a data protection review before it was built.

8.5 Contribution to the Sustainable Development Goals

The component contributes most plausibly to Goal 8 of the 2030 Agenda [11], decent work and economic growth, through target 8.2 on productivity through technological upgrading. The mechanism is narrow but real: a judgement that currently requires years of experience is made available to a grower who does not yet have them.

A secondary contribution to Goal 12 arises from avoided waste. A pollination attempt on an unfit plant consumes nine months of a plant's productive capacity and produces nothing. The claim defensible on present evidence is that the component changes which plants are selected; the resulting reduction in wasted attempts across a season has not been measured in a working shade house and is not claimed.

A possible negative effect deserves naming. A system that makes pollination decisions easier could increase the total number of attempts, and therefore the resources consumed by unsuccessful ones, if its recommendations are weaker than a skilled grower's own judgement. On the present margin of 0.078 over baseline that risk cannot be dismissed, and it is a further argument for presenting the system as advisory rather than authoritative.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis set out to determine how much of the Vanda pollination decision can be taken from a photograph, and what can honestly be said about a proposed cross from a register that records only successes. Both questions now have measured answers.

Readiness is predicted at 0.717 accuracy and 0.715 macro F1 against a 0.639 majority-class baseline, out of fold, on a split grouped by physical plant, with no trait declared by the grower at inference time. The Not Suitable class, whose misclassification is the expensive error, is recovered at 98 per cent. A two-stage input gate refuses 91.9 per cent of non-orchid photographs including every one of the 309 images across seven non-plant categories, at an area under the curve of 0.921 and a false-refusal rate of 2.2 per cent on in-domain photographs.

Cross compatibility is reported as an evidence tier over registered precedents and never as a probability. That restraint is the thesis's clearest conclusion about the data rather than about the method: a register of survivors has no denominator, and a percentage derived from it would be survivorship bias presented as a measurement.

The most transferable result is methodological. The same model on the same data scored 1.000 under an image-level split and 0.717 under a split grouped by plant. A protocol that detects a defect in its own author's favour is better evidence of its worth than one that merely produces good numbers.

9.2 Achievement of objectives

Table 9.1 maps each objective stated in Section 1.5 to its outcome and to the evidence that settles it.

Table 9.1: Achievement of the research objectives

Objective	Outcome	Evidence
O1 Readiness from one photograph, no declared traits	Achieved	0.717 against 0.639 baseline, out of fold, grouped by plant (Section 6.1)
O2 Refuse non-orchid photographs before any verdict	Achieved	91.9% refused, AUC 0.921, 2.2% false refusals (Section 6.2)
O3 Resolve cross compatibility from the registered record	Achieved in part	Four-tier scheme working over 27 crosses; coverage is the limitation (Section 6.2.3)

Objective	Outcome	Evidence
O4 Validate against a baseline on a deployment-realistic split	Achieved	Grouped protocol; leakage detected and corrected (Sections 3.5, 6.1)
O5 Integrate without duplicating another component's model	Achieved	Duplication measured at 0.522 AUC and rejected; provider contract used (Section 6.4)

Four objectives are achieved and one in part. O3 is reported as partial because the mechanism works correctly over a knowledge base of 27 crosses, which means a grower naming two parents outside it most often receives the no-precedent tier. The tier is the honest answer; it is not yet a useful one often enough.

9.3 Individual research and product contributions

The research contributions are the measured readiness figure on a protocol that does not flatter itself; the demonstration that a one-class gate fitted on in-domain photographs alone generalises to unseen out-of-domain categories; the documented leakage mechanism and its magnitude; and the negative results establishing that the orchid register cannot support a success probability and that disease is not readable from a whole-plant photograph.

The product contributions are the photograph-based readiness assessment with visible trait provenance; the precedent lookup with direction advice; the refusal behaviour that has since become the architectural pattern for optional models across the integrated system; and the capability-based availability check now used for every optional dependency in the backend.

The full statement of which work in the project is the author's own, and which is relied upon from other components, is given in Section 1.8.

9.4 Future directions

Five pieces of work are prioritised by the strength of evidence they would add rather than by the effort they would take.

First, the data the grower logbook was built to collect. The instrument itself has since been implemented and is described in Section 5.2: growers record an attempted cross with both parents photographed, the covariates that distinguish a genetically

incompatible pairing from a mistimed or poorly executed one, and the stage at which the attempt stopped. What remains is the thing no amount of engineering supplies, which is a season of entries. The feedback interval is the obstacle: a Vanda capsule takes about nine months, so the outcome that validates this component's own readiness model cannot be observed inside the period of this project. Until enough attempts have accumulated the store reports counts with their denominators and suppresses rates below a minimum sample size, and that behaviour should be retained rather than relaxed as the first few records arrive.

Second, enlargement of the Moderate class. Two plants cannot define a class, and this is a limitation of sample size rather than of method; a further season of photography addresses it directly.

Third, an independent second annotator and a measured inter-rater agreement, which would convert the construct-validity threat of Section 7.5 from an acknowledged weakness into a quantified one.

Fourth, regression coverage of this component's own interface contracts. The repository gained an automated suite during the project and it has already proved its worth on this component, rejecting a change that reached the shared database outside the agreed chokepoint - a defect no behavioural test would have caught, because the call succeeded. What it covers are project-wide invariants. The contracts of Table 5.2 are verified but not protected, and the component's own history shows how a silent fallback can survive a passing check.

Fifth, a pollen-viability assessment from a close-up of the column and anther cap. The present component assesses whole-plant readiness, which is the right measure for the pod parent; the pollen parent needs only viable pollinia, and that is visible at close range but not in a whole-plant photograph. This would require new annotation and is honestly a separate piece of research.

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GLOSSARY

Term	Meaning
Grex	The collective name given to all offspring of a particular cross between two orchid parents, registered once when the hybrid is first flowered and named.
Pod parent	The plant that receives the pollen, forms the seed capsule and carries it to maturity. Named first by breeding convention.
Pollen parent	The plant that donates the pollinia. Its own vigour matters far less than the pod parent's, since it bears none of the cost of the capsule.
Pollinia	The waxy masses in which orchid pollen is packaged, transferred whole during hand pollination.
Nothogenus	The name given to a genus-level hybrid, that is, a cross between plants of two or more different genera.
Out-of-fold score	A score computed for a record by a model trained without that record, and in this work without any record from the same plant.
Majority-class baseline	The accuracy obtained by always predicting the most common class of a split. Here 0.639.
Keep fraction	The proportion of reference distances falling below a gate threshold, used to set that threshold by quantile rather than by hand.
Survivorship bias	The error of drawing conclusions from a record that contains only the cases that succeeded.
Leave-one-plant-out	A validation scheme in which each plant in turn is held out entirely and predicted by a model fitted on the remaining plants.

APPENDICES

The appendices carry the research instruments, the extended results and the measurements that the chapters refer to but do not reproduce, together with the academic integrity and AI use disclosure required by the module policy. Each is cited from the text.

Appendix A Annotation protocol and full feature list

Labelling followed the written rules below. Where a plant sat between two definitions the grower was consulted and the decision recorded against the plant identifier, so that every image of that plant carries the same label. The rules are reproduced here because readiness has no ground truth independent of expert judgement, and a reader is entitled to see the judgement that was applied.

Table A.1: Operational annotation rules for the three readiness classes

Class	Assign when	Do not assign when
Suitable	The plant is mature, the leaves are turgid and evenly coloured, no active lesion is visible, the root system is sound, and the plant is in flower or in bud	Any active lesion is present, or the plant has not reached flowering size
Moderate	The plant is usable but compromised: minor leaf damage, uneven vigour, or a recently resolved problem. It can donate pollen but should not be asked to carry a capsule this season	The compromise is severe enough that the plant should not be bred from at all
Not Suitable	The plant is immature, visibly stressed, or shows active disease; or it has recently carried a capsule and has not recovered	The only defect is cosmetic damage to an older leaf with no active margin

The four categorical traits recorded alongside the class are leaf condition (healthy, moderate, weak), plant strength (strong, moderate, weak), disease visible (yes, no) and flower condition (flowering, budding, none). At training time these are the annotator's values; at inference time they are measured or supplied, as Section 3.3.2 describes, and never required from the grower.

Table A.2 names all 66 features in the descriptor. They are computed over the segmented region rather than the whole frame, for the reason given in Section 3.3.1.

Table A.2: The 66 features by group, named in full

Group	Features
Colour statistics (12)	mean_r, mean_g, mean_b, mean_h, mean_s, mean_v, std_r, std_g, std_b, std_h, std_s, std_v
Colour histograms (24)	hist_h_0 to hist_h_7, hist_s_0 to hist_s_7, hist_v_0 to hist_v_7
Contour shape (6)	contour_area, contour_perimeter, aspect_ratio, circularity, solidity, contour_count
Gabor texture (16)	gabor_mean and gabor_std at four orientations by two scales: 0_0, 0_1, 1_0, 1_1, 2_0, 2_1, 3_0, 3_1
Vegetation ratios (4)	green_ratio, brown_ratio, green_brown_ratio, green_pixel_count
Encoded traits (4)	leaf_condition_encoded, plant_strength_encoded, disease_visible_encoded, flower_condition_encoded

Appendix B Input gate validation record

The stage 1 gate was measured on 543 photographs assembled from sources never used in fitting: 455 that are not orchids and 88 that are. The row-level record is held as a CSV with one line per image, so any individual decision can be re-examined rather than taken from the summary in Table 6.3. Table A.3 gives its schema.

Table A.3: Schema of the gate validation record, 543 rows

Column	Contents
path	Location of the image within the validation set
label	Ground truth: orchid or not_orchid
category	One of the thirteen categories in Table 6.3
distance	Measured cosine distance to the tenth nearest reference neighbour
accepted	The gate's decision at the 0.25 threshold

Two rows in that file deserve comment because they were reclassified during collection rather than at analysis time. Both were gathered under an orchid search term and both are depictions rather than plants: one is a moulded plastic refrigerator magnet in the shape of an orchid flower, the other a nineteenth-century botanical lithograph. A system that assesses whether a plant is fit to breed from should refuse both, so they were moved to a depictions category and the reason recorded in the dataset README. The gate refuses both.

The headline figures from this record are 418 of 455 non-orchid images refused, an area under the ROC curve of 0.921 for orchid against non-orchid by distance, and 58 of 88 unseen web orchid photographs accepted. Grouped five-fold cross-validation

over the 1,190 project photographs gives a false-refusal rate of 2.2 per cent on images of the kind the gate was fitted on, which is the figure that describes deployment.

Appendix C Measured trait predictors

Section 3.3.2 states that the four categorical traits are measured from the image at inference time rather than demanded from the grower. Two of them, leaf condition and plant strength, are predicted by their own classifiers; this appendix reports how well, because the claim that traits are measured is only worth as much as those measurements.

Both predictors were evaluated under StratifiedGroupKFold grouped by plant, on the same 357 photographs from 28 plants, with three feature sets compared: the handcrafted descriptor, the frozen CNN embedding, and the two combined. The embedding won for both traits. Table A.4 reports the outcome.

Table A.4: Measured performance of the two trait predictors, out of fold and grouped by plant

Trait	Best model	Features	Accuracy	Balanced accuracy	Baseline
Plant strength	Extra Trees	CNN embedding	0.698	0.514	0.546
Leaf condition	Extra Trees	CNN embedding	0.557	0.425	0.501

Neither result is strong and both are reported as they stand. Plant strength clears its baseline by 0.152, which is a usable margin. Leaf condition clears its baseline by 0.056, which is slight; its balanced accuracy of 0.425 across three classes shows that the weak class, which has four plants behind it, is largely missed. The practical consequence is that a leaf-condition value measured from the image carries less weight than one the grower has corrected, which is exactly why the trait-resolution block reports the origin of every value and why an unresolved trait damps the confidence rather than being silently defaulted.

Appendix D Evidence that disease cannot be read from a whole-plant photograph

Section 5.3 explains that this component consumes Component 2's disease classifier rather than training one of its own, and Section 6.4 reports the integration. That decision was taken on evidence, not on grounds of tidiness, and the evidence is recorded here.

A disease signal was probed directly from the component's own whole-plant photographs. Candidate descriptors were computed over the segmented region and ranked by their individual ability to separate the ten diseased plants from the eighteen healthy ones. The four strongest are given in Table A.5.

Table A.5: The four most informative disease descriptors measured on whole-plant photographs

Descriptor	Individual ROC AUC	What it measures
dark_rel_70	0.700	Fraction of the plant darker than the 70th percentile
lesion_n_k25_c40	0.672	Count of candidate lesion blobs after clustering
val_mean	0.656	Mean value channel over the segmented region
lab_b_mean	0.650	Mean b channel in CIE Lab, the blue-yellow axis

Combining the descriptors into a classifier and validating it leave-one-plant-out produced the result in Table A.6. The honest reading is that it does not work. The area under the curve of 0.522 is barely distinguishable from the 0.5 of a coin, and the leave-one-plant-out accuracy of 0.571 is below the 0.643 obtained by answering healthy for every plant. A model that is worse than the majority class is not a weak model; it is evidence that the signal is not present in this input.

Table A.6: Disease prediction from whole-plant photographs, measured leave-one-plant-out

Measure	Value	Reference
Plants	28 (10 diseased, 18 healthy)	-
ROC AUC	0.522	0.500 is chance
Leave-one-plant-out accuracy	0.571	0.643 majority baseline
Leave-one-plant-out F1	0.500	precision 0.429, recall 0.600
In-sample accuracy at fitted threshold	0.643	equal to the baseline

The reason is not mysterious. A lesion occupies a few square centimetres of one leaf; a whole-plant photograph taken at arm's length renders that as a handful of pixels, and the descriptors that respond to it respond just as readily to shadow, to a wet leaf, or to the dark interior of the plant. Component 2 photographs a single leaf at close range on 28,782 training images, which is the right input and roughly eighty times the data.

This is why the provider interface returns a confidence of zero when Component 2 is unavailable, rather than falling back to the descriptors above. A zero confidence is an honest statement that the question was not answered. A fallback built on a 0.522 area under the curve would be a statement that it was.

Appendix E Registered cross knowledge base

The second level's knowledge base holds 27 crosses drawn from the International Orchid Register [3], listed in full below. Fourteen are interspecific, between two species of Vanda, and thirteen are intergeneric, involving one of Aerides, Aeridovanda, Arachnis, Aranda, Luisia, Papilionanthe, Paraphalaenopsis or Renanopsis. The seed parent is given first throughout, following the convention described in Section 3.3.6. Table A.7 lists them.

Two properties of this table bear on how its contents may be used. It contains only crosses that succeeded and were registered, which is the basis of the evidence tiering in Section 6.2.3 and of the refusal to report a success rate. And V. tessellata appears in fifteen of the 27 rows, which reflects both its availability in Sri Lanka and its documented willingness to cross outside its own genus; it is the rescue parent the component suggests when an imported hybrid pairing sets no pod.

Table A.7: The 27 registered crosses held in the knowledge base, seed parent first

Seed parent	Pollen parent	Resulting grex	Year
V. coerulea	V. sanderiana	V. Rothschildiana	1931
V. Rothschildiana	V. sanderiana	V. Onomea	-
V. luzonica	V. sanderiana	V. Manila	-
V. merrillii	V. sanderiana	V. Mary Foster	-
V. dearei	V. sanderiana	V. Ellen Noa	-
V. sanderiana	V. denisoniana	V. Denisander	-
V. tessellata	V. denisoniana	V. Golden Doubloon	-
V. tessellata	V. testacea	V. Stephen-Scott Young	-

Seed parent	Pollen parent	Resulting grex	Year
V. coerulea	V. tessellata	V. Violeta	1959
V. coerulea	V. tessellata	V. x amoena	-
V. sanderiana	V. tricolor	V. Tatzeri	1919
V. sanderiana	V. suavis	V. Burgeffii	-
V. tessellata	V. Fat Tuesday	V. Alexander Bristol	2026
V. tessellata	V. wightii	V. Allison Craig Scanlan	2021
V. tessellata	Aerides lawrenceae	Aeridovanda Frank Johnston	1963
V. tessellata	Aerides leeana	Aeridovanda Les Miller	1962
V. tessellata	Aranda Redland Sunset	Aranda Michael Coronado	1993
V. tessellata	Luisia macrantha	Luisanda Sarojini Menon	2018
V. tessellata	Papilionanthe Poepoe	Papilionanda Rosanna	2004
V. tessellata	Paraphalaenopsis denevei	Paravanda Genevieve Tan	1970
V. tessellata	Renanopsis Lena Rowold	Renvanvanda Sian Hee Wee	1978
Aerides flabellata	V. tessellata	Aeridovanda Happiness	2025
Aerides maculosa	V. tessellata	Aeridovanda Mem. Nilkanth Bhide	2023
Aerides odorata	V. tessellata	Aeridovanda Mem. Premier Chou	1981
Aeridovanda Suwannaphum Sun	V. tessellata	Aeridovanda Suwannaphum Willow	2025
Arachnis flos-aeris	V. tessellata	Aranda Bussaracum	1975
Aranda Blue Star	V. tessellata	Aranda Darwin Peh	1982

Appendix F Interface and data contract specification

This appendix records the request and response shapes of the component's five endpoints, so that the integration claims of Chapters 4 and 5 can be checked without reading the source. All five sit under the shared prefix /api/v1/pollination. Table A.8 gives the contracts and Table A.9 is a field dictionary for the responses.

Table A.8: Endpoints exposed by the component, with their request and response contracts

Endpoint and method	Request	Response
POST /assess	multipart/form-data: image (required), leaf_closeup (optional), and any of leaf_condition, plant_strength, disease_visible, flower_condition that the grower has explicitly corrected. Unanswered traits are omitted from the request entirely	suitability (Suitable | Moderate | Not Suitable), confidence, trait_resolution (value and origin for each trait), input_check (distance, threshold, vegetation fraction, orchid probability), guidance text

Endpoint and method	Request	Response
POST /compatibility	application/json: pod_parent, pollen_parent, and optionally pod_health and pollen_health carrying a readiness result for either plant	tier, tier_label, class, precedents (seed parent, pollen parent, grex, year), warnings, direction advice, expected_offspring where both parents are documented species. No success probability is returned under any circumstances
POST /compatibility/rank	application/json: pod_parent and a list of candidate pollen donors	The candidates ordered by evidence tier, strongest first, each with its tier and precedent count
GET /compatibility/parents	none	The parent names the knowledge base recognises, for type-ahead in the application. A name absent from this list can still be assessed; it falls back to genus-level evidence
GET /health	none	Which models are loaded, so a partially provisioned server is visible rather than silent

Table A.9: Principal response fields and their meanings

Field	Type	Meaning
suitability	string	One of Suitable, Moderate, Not Suitable
confidence	float 0-1	Classifier probability, damped when the input sits far from the training distribution
trait_resolution	object	For each of the four traits: the resolved value and its origin (grower, measured, provider, unresolved)
input_check.distance	float	Cosine distance to the tenth nearest reference neighbour
input_check.threshold	float	The gate threshold in force, 0.25
input_check.vegetation	float 0-1	Segmented vegetation fraction; below 0.02 the image is refused
input_check.orchid_probability	float 0-1	Stage 2 filter output
tier	string	registered | genus_proven | undemonstrated | blocked
precedents	array	Registered crosses supporting the tier, each with seed parent, pollen parent, grex and year

Appendix G Verification evidence and research artefact inventory

Verification of this component rests on measured validation sets written to artefacts, together with the documented interface checks listed in Table 5.2. There is no automated regression suite for the component at the time of submission; the interface behaviours in Table 5.2 were confirmed by hand against a running backend and are

recorded as such rather than reported as a pass rate. Converting them into automated tests is named as outstanding work in Section 9.4.

What does exist is a complete record of every measurement quoted in this thesis. Table A.10 lists each artefact, the file that holds it and the claim it supports, so that no figure in Chapter 6 rests on a number that cannot be traced to a file.

Table A.10: Research artefacts, their contents and the claims they support

Artefact	Contents	Supports
training_results.json	Evaluation method, fold count, grouping variable, majority baseline, per-model metrics and confusion matrices, dataset composition, explicit caveats	Tables 6.1 and 6.2; Figures 6.1 and 6.2
orchid_gate.txt	Reference set size, threshold and its keep fraction, distance quantiles, cross-validated false-refusal rate, per-category validation results	Table 6.3 and Figure 6.3
orchid_gate_validation.csv	543 rows: path, label, category, distance, accept decision	Appendix B; any individual gate decision
flower_filter_cv.txt	Positive and negative counts, threshold, third-source transfer test, per-category results	Section 6.2.2
flower_model_results.json	Cut-out model cross-validated metrics and the two-direction transfer test	Figure 6.4, the retained negative result
trait_training_results.json	Three feature sets compared for each trait, per-model metrics, baselines	Appendix C
disease_validation.json	Leave-one-plant-out disease metrics and the fitted threshold	Appendix D, Table A.6; Section 6.4
disease_feature_ranking.csv	Individual ROC AUC of each candidate disease descriptor	Appendix D, Table A.5
classification_reports.txt	Per-class precision, recall and F1 for every candidate classifier	Section 6.1
registered_crosses.csv	The 27-row knowledge base with genera, cross type, year and source	Appendix E, Section 6.2.3
models/ (11 files)	Serialised gate, flower filter, trait predictors, four candidate classifiers and preprocessors	Reproduction of any reported figure

Every stochastic component uses a fixed random state of 42, and the dependency versions are pinned in the shared requirements file. Reproducing any figure in Chapter 6 requires only the dataset, those pins and the training entry point for the stage concerned.

Appendix H Academic integrity and AI use disclosure

This appendix discloses the use of generative artificial intelligence and other automated tools in the preparation of this thesis and of the component it describes, in accordance with the module policy. The declaration on page i applies in full: the research, the data, the models, the measurements and the conclusions are the author's own, and the author remains responsible for the accuracy of everything reported here regardless of the tools used to prepare it. Table A.11 records tool use by activity, with the verification carried out in each case.

Table A.11: Disclosure of tool use by activity, with the verification carried out in each case

Activity	Tool and purpose	Verification performed
Dataset collection and annotation	No AI tool used. Photographs taken by the author; labels assigned by the author against the written rules in Appendix A with the grower consulted on ambiguous cases	Not applicable
Model design, training and evaluation	Conventional libraries only (scikit-learn, XGBoost, PyTorch). No generative AI chose a model, a hyperparameter or a threshold	Every reported figure re-read from the saved artefacts listed in Table A.10
Source code	A generative AI assistant was used for code review, debugging and refactoring suggestions during development of the backend and application	All code was read, run and tested by the author; defects found by the author are recorded in Sections 4.4, 5.2, 5.3 and 5.5
Figures in this thesis	A generative AI assistant was used to write the plotting scripts that render Figures 1.1 to 6.4 from the saved result files	Each figure was compared against the artefact it plots; the numbers are read from file at build time, not transcribed
Literature search	Conventional database search (ACM, IEEE). AI assistance was not used to generate citations	Every reference in the list was located and read; no reference is cited that the author has not seen

Two points of principle are recorded because they bear on integrity rather than merely on process. First, no result in this thesis was generated, estimated or completed by a language model. Every number in Chapter 6 and in the appendices is read from a file written by a training or validation script, and Table A.10 maps each file to the claim it supports precisely so that this can be checked. Second, the negative results were retained deliberately: the cut-out flower model that scored 0.999 in cross-validation and 0 of 38 on real photographs, the disease probe that fell

below its own majority baseline, and the image-level split that reported a perfect 1.000. A thesis prepared with AI assistance is under a particular obligation to show its failures, because fluent prose makes unearned claims easier to write and harder to spot.